Estimation and Inference on Two-Threshold Value Functionals*

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Version 4

Abstract

We study estimation and inference for value functionals of the form $\Theta_0 = \mathbb{E}\big[\mathbb{1}\{g_0(X) \geq 0\}\,\mathbb{1}\{h_0(X) \geq 0\}\,v_0(X)\big]$, where g_0 and h_0 are two nonparametric contrast functions estimated from the same experiment. Such *two-threshold* functionals arise whenever a planner imposes a pointwise admissibility screen on top of a benefit margin: leading examples are the share of the population that benefits in the short run but is harmed in the long run under a surrogate-index treatment rule (Athey et al., 2024), and treatment-path welfare components of an optimal dynamic regime.

Our primary contribution is the *second-order* shape calculus of such functionals, which we derive in full and verify numerically. The first pathwise derivative is a sum of two boundary (Hausdorff) integrals over the two margins. The second contains, in addition to two margin curvature forms, a codimension-two *corner* integral over $\{g_0 = 0\} \cap \{h_0 = 0\}$ together with two corner *diagonal* terms weighted by the cosine of the intersection angle. We then give a complete order accounting of the terms this generates in a sieve plug-in expansion: the corner contributes $O(K_n/n)$, the margins $O(K_n^{1+1/d}/n)$, and and we show that the margin bound, unlike the corner bound, is exhibited as attained only in mesh-resonant examples, its exact order in general remaining open. The accounting implies that margin-by-margin quadratic debiasing à la Chen and Gao (2026) leaves a corner cross term that is rate-relevant on the window $(d+1)/2 < s \leq d-1$, nonempty for $d \geq 4$; we therefore construct corner-aware

*This section extends the single-threshold framework of Chen et al. (2025) and its debiased-estimation companion to value functionals defined by the intersection of two estimated decision boundaries. It is written to be self-contained; notation follows the JMP draft `may_2026.tex`.

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split-sample and leave-one-out corrections, with a tuning-free half-sample jackknife representation.

The inferential results are stated as *conditional* theorems and we are explicit about what is and is not proved. Under a third-order remainder bound we do not establish, the corner-aware split-sample estimator attains $n^{-s/(2s+1)}$ under $s > (d+1)/2$; under the crude bound available to us the range shrinks to $s > 3(d+1)/4$. A centered normal limit requires strict undersmoothing and, for the plug-in, strict $s > d$; we give the explicit sieve-dimension windows. We also correct an error in the previous version's *doubly debiased* estimator, which combined the quadratic correction with a cross-fitted Riesz correction: as constructed there, the Riesz score reintroduces exactly the own-half second-order diagonals the quadratic correction removes. We derive the score of the composite split-sample functional, which is $D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^A)$ rather than $\frac{1}{2}D\Theta(\hat{\gamma}^A)$, and give a three-fold construction that repairs it. Monte Carlo experiments calibrated to the Banerjee et al. (2015) graduation experiment, following the Wasserstein-GAN calibration protocol of Chen and Ritzwoller (2023), illustrate the corner mechanism; their doubly debiased cells evaluate the superseded construction and we read them accordingly.

Status of the results. Theorem 3.1 (the second shape derivative), Proposition 2.1, the polarization identity (37), the support-edge term of Remark 1.10, and the jackknife finite-difference identity of Lemma 4.2 are established, and verified numerically to ten significant digits (Appendix C). The order accounting of Section 3.2 is a set of *upper bounds* whose exponent arithmetic we have checked symbolically; Remark 3.4 identifies where those bounds are and are not attained. Theorems 2.2, 4.6, 5.3 and 5.8 are *conditional*: each displays, in one place, every remainder condition it uses — Assumption 1.8 collects them — and Appendix B, §B.4 lists exactly which of those we have not verified from primitive conditions. The reader should treat the four theorems as specifications of what must be proved, complete as to their hypotheses but not as to their proofs.

A drafting convention that matters for reading Section 4: we distinguish throughout between the *ideal* split-sample estimators, built from the closed-form $D^2\Theta$ of Theorem 3.1, and their *feasible* finite-difference (jackknife) counterparts. The decomposition identities hold exactly *within* each family and only up to a fourth-order error *across* them; v3 conflated the two.

Revision note (v4). This version repairs internal defects of v3 that were mathematical, not merely unproved. *(i)* v3 asserted $\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\mathrm{mar}} - \widehat{\mathrm{corner}}$ with $\hat{\Theta}_{SS}$ the ideal derivative-based estimator. That is false: the polarization identity holds exactly for the *jackknife* family, and across families there is a fourth-order gap. Section 4 now carries both families with their own exact identities ((34) and (36)). *(ii)* Theorem 2.2(i) controlled only the linear and quadratic terms while the expansion contains a third-order remainder; Assumption 1.8 now collects every remainder condition used anywhere in the paper, and each theorem cites it. *(iii)* The jackknife centred CLT needs its own sieve window $a < 3d/(4d+7)$, not merely the smoothness condition (33); Theorem 4.6(iii) supplies it. *(iv)* The second-moment "exact sandwich" of v3 was written for a single contrast regression; the paper's estimator runs *four separate arm regressions*, so the exact expression is an arm-wise sum of sandwiches (23), the localization error is additive rather than relative, and the $h_0 = -\tau_Y$ convention flips a sign. *(v)* Assumption 5.1(c) claimed Hausdorff-distance plus region-measure convergence was "equivalent" to normal-graph convergence. It is not — a surface can be Hausdorff-close while oscillating, with divergent surface measure — so (c) now *requires* a normal-graph representation with C^1 control, plus a convention for empty margins. *(vi)* The SS expansion also carries an evaluation-point remainder $\{D^2\Theta(\bar\gamma) - D^2\Theta(\gamma_0)\}[\Delta,\Delta]$, distinct from the Taylor remainder; both are now controlled by a single *second-derivative modulus* condition, Assumption 1.8(a), which has the further advantage of not presupposing that $D^3\Theta$ exists — the segment from γ_0 to $\bar\gamma$ passes through the truth, which is only C^2. *(vii)* v3's proof of Theorem 4.6 justified variance consistency by claiming the correction is $o_p(\mathrm{se})$. In the SS smoothness window it need not be; consistency follows instead from the asymptotically linear representation. *(viii)* The corner noncancellation condition was imposed on population coefficients, which does not prevent the spline-weighted sequence from cancelling; it is now imposed on the actual normalized sequence $\frac{n}{K_n}\mathcal{B}_{\mathcal{C},n}$, and the full-bracket and cross-term conditions are stated separately, since neither implies the other. *(ix)* We add a condition that no earlier version or critique identified: in the empirical-measure case the plug-in carries a *stochastic-equicontinuity* term $\frac{1}{\sqrt{n}}\mathbb{G}_n[m_{\hat\gamma} - m_{\gamma_0}]$ of order $(K_n/n)^{3/4}$, which imposes $a < d/(3d-2)$ and binds for $d \geq 3$ (Remark 1.9). *(x)* Assorted repairs: uniform overlap; the stacked standard-deviation norm defined on the summed influence rather than as a scalar; the bilinear extension of $D^2\Theta$ stated by polarization; "shape (Gâteaux) derivative" replacing "pathwise derivative"; arm-residual consistency added to Proposition 2.4; the composite representer conditions and the empirical-measure fold convention added to Section 5.2; $\asymp$ demoted to O in Proposition 4.10; the LOO claim softened; and the internal contradiction between the abstract, the main text, and Appendix B.1 over mesh resonance resolved in favour of the upper-bound reading.

Relative to v1–v2 the corrections recorded in the v3 note stand: the invalidity of the old doubly debiased estimator and its composite-score repair, the separation of rate statements from centred CLTs, the margin/corner order distinction, and the two incorrect claims of

Remarks 5.4 and 5.5. The Monte Carlo tables are reproduced unchanged from v1; Section 6.5 states what they do and do not support.

1 Setup and Examples

1.1 Data and estimands

The observed data are an i.i.d. sample $\{Z_i = (X_i, W_i, S_i, Y_i)\}_{i=1}^n$, where $X_i \in \mathcal{X} \subset \mathbb{R}^d$ are covariates with density f_0 on a bounded rectangular support, $W_i \in \{0, 1\}$ is a binary treatment with $(S_i(0), S_i(1), Y_i(0), Y_i(1)) \perp W_i \mid X_i$ and overlap $p_0(x) := \mathbb{E}[W_i \mid X_i = x] \in (0, 1)$, S_i is a short-run outcome (or surrogate), and Y_i is a long-run outcome. Define the arm means and contrasts

$$\mu_{S,0}(x, w) := \mathbb{E}[S_i \mid X_i = x, W_i = w], \qquad \tau_S(x) := \mu_{S,0}(x, 1) - \mu_{S,0}(x, 0), \tag{1}$$

$$\mu_{Y,0}(x, w) := \mathbb{E}[Y_i \mid X_i = x, W_i = w], \qquad \tau_Y(x) := \mu_{Y,0}(x, 1) - \mu_{Y,0}(x, 0). \tag{2}$$

Throughout we write $\gamma_0 := (g_0, h_0)$ for a generic pair of identified contrast functions built from $(\mu_{S,0}, \mu_{Y,0})$ — in the leading example $g_0 = \tau_S$ and $h_0 = -\tau_Y$ — and consider the *two-threshold value functional*

$$\Theta(\gamma_0) := \int_{\mathcal{X}} \mathbb{1}\{g_0(x) \geq 0\}\, \mathbb{1}\{h_0(x) \geq 0\}\, v_0(x)\, f(x)\, \mathrm{d}x, \tag{3}$$

where v_0 is a known, user-chosen value weight and f is the covariate density of the target population. As in the single-threshold analysis we distinguish the *known-density* case (f given, integral computed numerically) and the *empirical* case ($f = f_0$ unknown, integral replaced by the sample average $n^{-1} \sum_i \mathbb{1}\{\cdot\} v_0(X_i)$); the first-order asymptotic theory is identical in both, because the additional sampling error of the empirical measure is $O_p(n^{-1/2})$, which is negligible relative to the slower boundary-driven rate of (3). It is *not* negligible at the sample sizes of Section 6, and we therefore carry it explicitly in the variance formula (15) below.

1.2 Leading examples

Example 1.1 (Surrogate-induced harm share)**.** *A planner who observes only the short-run outcome adopts the surrogate-index rule "treat iff $\tau_S(x) \geq 0$." The long-run criterion would* *instead require $\tau_Y(x) \geq 0$. With $g_0 = \tau_S$, $h_0 = -\tau_Y$, and $v_0 \equiv 1$, (3) becomes*

$$\theta_0 = \mathbb{P}_f\big(\tau_S(X) \geq 0,\ \tau_Y(X) \leq 0\big),$$

the mass of the population that the surrogate rule treats but that is harmed in the long run — a direct diagnostic of the surrogate-index approach of Athey et al. (2024). This is the functional we take to the calibrated Monte Carlo of Section 6.

Example 1.2 (Admissibility-constrained policy value)**.** *In the two-item-checklist planner problem of the JMP draft, a hard admissibility screen $g_0(x) \geq 0$ (e.g. no expected severe side effect) is imposed before the benefit margin $h_0(x) \geq 0$; treated-population aggregates take the form (3) with v_0 equal to a utility, cost, or eligibility weight.*

Example 1.3 (Dynamic treatment-path welfare)**.** *The $(1,1)$ path-welfare component of an optimal two-stage dynamic regime, $V^*_{11} = \mathbb{E}[Y^{(1,1)}\mathbb{1}\{\kappa(S) \geq 0\}\mathbb{1}\{\delta(X^{(1)}) \geq 0\}]$, has the same two-threshold structure, with the additional complication that the first-stage contrast κ is itself a functional of the second-stage threshold. The static analysis of this section is the template for that case; the extra "moving-boundary-within-a-moving-boundary" terms are treated in the DTR section of the draft.*

1.3 Geometry and maintained assumptions

Write $w(x) := v_0(x)f(x)$ for the boundary weight, and define the two *margins* and the *corner*

$$\begin{aligned} \mathcal{M}_g &:= \{x : g_0(x) = 0,\ h_0(x) > 0\}, \qquad \mathcal{M}_h := \{x : h_0(x) = 0,\ g_0(x) > 0\}, \\ \mathcal{C} &:= \{x : g_0(x) = h_0(x) = 0\}. \end{aligned} \tag{4}$$

Assumption 1.4 (Model)**.** *(a) $\{Z_i\}$ is i.i.d.; $\mathcal{X}$ is bounded rectangular; and overlap is* uniform, *$p_0(x) \in [\underline{p}, 1-\underline{p}]$ for some $\underline{p} > 0$ — pointwise $p_0(x) \in (0,1)$ does not suffice, because the inverse-propensity weights of (25) and the arm-wise Gram matrices R_w of (23) must be bounded uniformly; (b) $\mu_{S,0}(\cdot, w), \mu_{Y,0}(\cdot, w) \in \Lambda^s(\mathcal{X})$ (Hölder class) with $s > 1$, for $w \in \{0,1\}$; (c) the conditional second moments $\sigma^2_S(x,w), \sigma^2_Y(x,w)$ of the regression residuals $\epsilon_{S,i} := S_i - \mu_{S,0}(X_i, W_i)$, $\epsilon_{Y,i} := Y_i - \mu_{Y,0}(X_i, W_i)$ are bounded and bounded away from zero, the conditional correlation $\rho_{SY}(x,w) := \operatorname{Corr}(\epsilon_{S,i}, \epsilon_{Y,i} \mid X_i = x, W_i = w)$ is well defined, and $\mathbb{E}[|\epsilon_{o,i}|^{2+\varsigma} \mid X_i, W_i]$ is bounded for some $\varsigma > 0$ and $o \in \{S, Y\}$; (d) v_0 is bounded and continuously differentiable, and f is bounded with bounded gradient, absolutely continuous w.r.t. f_0 with bounded Radon–Nikodym derivative bounded away from zero on a neighborhood of $\mathcal{M}_g \cup \mathcal{M}_h$.*

Assumption 1.5 (Regular margins and transversal corner)**.** *g_0 and h_0 are twice continuously differentiable in neighborhoods of $\{g_0 = 0\}$ and $\{h_0 = 0\}$ respectively, with (a) $\|\nabla g_0\| \geq \underline{c} > 0$ on $\{g_0 = 0\}$ and $\|\nabla h_0\| \geq \underline{c} > 0$ on $\{h_0 = 0\}$; (b) (*transversality*) on $\mathcal{C}$, the angle $\omega(x)$ between $\nabla g_0(x)$ and $\nabla h_0(x)$, defined by $\cos \omega(x) = \langle \nabla g_0, \nabla h_0 \rangle / (\|\nabla g_0\| \, \|\nabla h_0\|)$, satisfies $|\sin \omega(x)| \geq \underline{c}_T > 0$; (c) $\mathcal{H}^{d-1}(\mathcal{M}_g) + \mathcal{H}^{d-1}(\mathcal{M}_h) < \infty$ and $\mathcal{H}^{d-2}(\mathcal{C}) < \infty$; (d) (*no support edge*) the level sets $\{g_0 = 0\}$ and $\{h_0 = 0\}$ meet $\partial \mathcal{X}$ only where w vanishes, so that no margin terminates at the edge of the covariate support and the boundary calculus below involves only the margins and their common corner.*

Under Assumption 1.5, $\mathcal{M}_g$ and $\mathcal{M}_h$ are $(d-1)$-dimensional C^2 submanifolds-with-boundary whose common relative boundary is the $(d-2)$-dimensional corner $\mathcal{C}$ (a finite set of points when $d = 2$; empty when $d = 1$ or when the constraints never bind jointly). Transversality rules out tangential intersections, at which the corner would inflate to a higher-dimensional set and the second-order calculus below would fail.

The remaining maintained conditions are nondegeneracy requirements. They are separated out because several statements below are two-sided ("$\asymp$", "rate-relevant", "minimax") and are false without them: a functional with $v_0 \equiv 0$, or with empty margins, or whose corner coefficient cancels exactly, satisfies Assumptions 1.4–1.5 and has no irregularity at all.

Assumption 1.6 (Parameter space and nondegeneracy)**.** *(a) (*Parameter space.*) $\Lambda_c^s(\mathcal{X})$ denotes the ball of radius c in the Hölder space of order s on $\mathcal{X}$, and the statistical model is*

$$\mathcal{P}_c^s := \Big\{ P : \mu_{S,0}(\cdot, w), \mu_{Y,0}(\cdot, w) \in \Lambda_c^s \text{ for } w \in \{0,1\}, \text{ and } P \text{ satisfies} \\ \text{Assumptions 1.4–1.5 and (b)–(c) below with common constants} \Big\}.$$

*Rates and minimax statements are uniform over $\mathcal{P}_c^s$. (b) (*Nonvanishing boundary weight.*)*

$$\mathcal{H}^{d-1}(\mathcal{M}_g) \vee \mathcal{H}^{d-1}(\mathcal{M}_h) \;\geq\; \underline{m} > 0 \qquad \text{and} \qquad \inf_{\mathcal{M}_g \cup \mathcal{M}_h} w \;\geq\; \underline{w} > 0.$$

At least one *margin must be nondegenerate; this is what makes Θ_0 irregular and, with the upper bound $\mathcal{H}^{d-1}(\mathcal{M}_g) + \mathcal{H}^{d-1}(\mathcal{M}_h) < \infty$ of Assumption 1.5(c), delivers $\sigma_n^2 \asymp K_n^{1/d}$. Without it the representer norm can be $O(1)$ and Θ_0 is $\sqrt{n}$-estimable. It is deliberately "$\vee$" and not "$\wedge$": the single-threshold submodel used for the minimax lower bound in Theorem 2.2(ii) has $\mathcal{M}_h = \emptyset$, and requiring both margins to be nondegenerate would exclude it from $\mathcal{P}_c^s$ and break the nesting argument. (c) (*Corner noncancellation.*) Let $\mathcal{B}_{\mathcal{C},n}$ and $\mathcal{B}_{\mathcal{C},n}^{\times}$ denote, respectively, the full corner coefficient (27) and its cross-term component (28) defined in Section 3.2; both* *are deterministic sequences depending on the sieve through the normalized second-moment functions $\mathcal{V}_{\bullet,n}/K_n$. Either $\mathcal{C} = \emptyset$, or*

$$\text{(c1)} \quad \liminf_n \left| \frac{n}{K_n} \mathcal{B}_{\mathcal{C},n} \right| > 0, \qquad \text{(c2)} \quad \liminf_n \left| \frac{n}{K_n} \mathcal{B}^{\times}_{\mathcal{C},n} \right| > 0. \tag{5}$$

(c1) is used for lower bounds on the plug-in's corner bias, (c2) for the bias that survives margins-only debiasing (Proposition 4.10). Neither implies the other, *and neither follows from pointwise nonvanishing of ϱ_{gh}: the bracket in (27) is signed, its three components may offset, contributions from different corner points may offset, and the spline-dependent weights $\mathcal{V}_{\bullet,n}(x)/K_n$ vary across corner points and with n. Both are needed only for lower bounds; no upper bound below uses either.*

Remark 1.7 ($\mathcal{C}$ is not automatically material)**.** *Assumption 1.6(c) is a genuine restriction, not a formality. At an orthogonal corner* ($\cos\omega = 0$) *with $\rho_{SY} = 0$ the bracket vanishes identically; with* $\mathrm{d} = 2$ *and several corner points the terms carry the sign of* $\cos\omega$ *at each point, which typically varies; and even with a fixed nonzero pointwise coefficient the n-dependent weights could conspire to cancel along a subsequence, which is why* (5) *is imposed on the normalized sequence rather than on population quantities. In the calibrated design of Section 6 the ten corners do not cancel (Table 4), but this is a property of that design.*

Assumption 1.8 (Remainders)**.** *Write $\bar{\Delta}$ for a generic first-stage error, η for a generic direction, and* $\mathrm{se}_n := \sqrt{K_n^{1/d}/n}$.

- *(a) (*Second-derivative modulus.*) There is a nondecreasing ϖ with $\varpi(0+) = 0$ such that, on a neighbourhood of γ_0 containing the estimated paths w.p. $\to 1$,*

$$\left|\{D^2\Theta(\gamma) - D^2\Theta(\gamma_0)\}[\eta, \eta]\right| \;\le\; \varpi\big(\|\gamma - \gamma_0\|_{C^1} \big) \, \|\eta\|_\infty \, \|\eta\|_{C^1} \,. \tag{6}$$

- *(b) (*Taylor remainder.*) With $R_3 := \frac{1}{2}\{D^2\Theta(\gamma_\xi) - D^2\Theta(\gamma_0)\}[\bar{\Delta}, \bar{\Delta}]$ the exact mean-value remainder of the second-order expansion* (17)*, and R_3^{SS} the total remainder of the corrected estimator defined in* (38)*, the relevant one is $O_p(r_n^*)$ where a rate is claimed and* $o_p(\mathrm{se}_n)$ *where a centred limit is claimed.*

- *(c) (*Stochastic equicontinuity; empirical-measure case only.*) With $m_\gamma(x) := \mathbb{1}\{g(x) \ge 0\}\mathbb{1}\{h(x) \ge 0\} v_0(x)$,*

$$n^{-1/2} \left|\mathbb{G}_n\big[m_{\hat{\gamma}} - m_{\gamma_0}\big]\right| = o_p(\mathrm{se}_n), \qquad \mathbb{G}_n := \sqrt{n}\,(\mathbb{P}_n - P). \tag{7}$$

- *(d) (*Lyapunov.*) For some $\varsigma > 0$, $n^{-\varsigma/2}\mathbb{E}\big[\,|v_S^*\epsilon_S + v_Y^*\epsilon_Y|^{2+\varsigma}\,\big]/\sigma_n^{2+\varsigma} \to 0$.*

Collecting the remainder conditions in one place is deliberate. Every theorem below cites Assumption 1.8 explicitly, so that no statement silently omits a term of its own expansion — a defect of v3's Theorem 2.2(i), which controlled the linear and quadratic terms of (17) but not R_3.

Three comments on the formulation. First, (a) deliberately avoids postulating a *third* shape derivative. The segment from γ_0 to $\bar{\gamma}$ passes through the truth, which Assumption 1.5 gives only as C^2; a third-order Taylor expansion around γ_0 would need C^3 there, contradicting the Hölder parameter space whenever $s < 3$. The mean-value form of (b) needs only D^2 plus the modulus (a). Second, (a) also controls the SS *evaluation-point* remainder: the estimator subtracts $D^2\Theta(\bar{\gamma})[\Delta, \Delta]$ whereas the cancellation argument uses $D^2\Theta(\gamma_0)$, and the difference is bounded by the same ϖ. Third, the strength of (b) is entirely inherited from ϖ; Remark 4.8 converts each candidate modulus into a smoothness requirement.

Remark 1.9 (The equicontinuity term, and why it is new)**.** *Condition* (7) *is not a technicality and was absent from v1–v3. Writing* $\hat{\Theta} - \Theta(\hat{\gamma}) = n^{-1/2}\mathbb{G}_n[m_{\gamma_0}] + n^{-1/2}\mathbb{G}_n[m_{\hat{\gamma}} - m_{\gamma_0}]$*, the first term is the* $O_p(n^{-1/2})$ *empirical-measure error carried explicitly in* (15)*. The second is a genuine empirical-process term that earlier versions absorbed into "*$O_p(n^{-1/2})$*" without argument. The class* $\{m_\gamma : \gamma \in \mathcal{S}^2_{K_n}\}$ *is VC of index* $O(K_n)$*, and* $\|m_{\hat{\gamma}} - m_{\gamma_0}\|^2_{L^2} \lesssim \|\hat{\gamma} - \gamma_0\|_\infty$ *because the two regions differ in a band of that width around the margins, so the standard maximal inequality gives*

$$n^{-1/2}\big|\mathbb{G}_n[m_{\hat{\gamma}} - m_{\gamma_0}]\big| = O_p\Big(\|\hat{\gamma} - \gamma_0\|^{1/2}_\infty \sqrt{K_n \log K_n/n}\Big) = \tilde{O}_p\big((K_n/n)^{3/4}\big), \tag{8}$$

using $\|\hat{\gamma} - \gamma_0\|_\infty \asymp \sqrt{K_n/n}$ *up to logs. This is* $o(\mathrm{se}_n)$ *iff*

$$a < \frac{d}{3d-2}, \qquad \textit{compatible with } a > \tfrac{d}{2s+1} \textit{ iff } s > \tfrac{3(d-1)}{2}. \tag{9}$$

At $d = 2$ *the edge is* $a < 1/2$*, which coincides with the off-diagonal edge* $d/(d+2)$ *and with* $s > 3/2 = (d+1)/2$*, so it does* not *bind in the designs of Section 6. For* $d \geq 3$ *it is the binding upper edge (*$3/7$ *at* $d = 3$ *against* $3/5$*), and it demands* $s > 3$ *rather than* $s > 2$*. Two caveats. The known-density case has no such term, the integral being computed against* f *rather than estimated. And* (8) *uses a global VC bound; a local-complexity argument exploiting that only the* $O(\varkappa^{-(d-1)})$ *basis functions meeting the margins are active could plausibly improve it, which we have not attempted. We state* (7) *as a condition and* (8) *as the sufficient bound we can currently supply.*

Remark 1.10 (Assumption 1.5(d) is substantive)**.** *Condition (d) is not a technicality. If a margin exits through* $\partial\mathcal{X}$ *at a point where* $w > 0$*, then* $\mathcal{M}_g$ *has a second kind of relative* *boundary* — a support edge — *and the second derivative* (18) *acquires an extra* $\mathcal{H}^{d-2}$ *integral over* $\{g_0 = 0\} \cap \partial\mathcal{X}$. *Because the support is* fixed *while the level set flows, only the "induced motion of the cut" mechanism (step (4) of Appendix A) operates: the edge contributes a* u^2 diagonal *term*

$$-\int_{\{g_0=0\}\cap\partial\mathcal{X}} \frac{u^2\, w\, \cos\omega_\partial}{\|\nabla g_0\|^2\, |\sin\omega_\partial|}\, \mathrm{d}\mathcal{H}^{d-2}, \qquad \cos\omega_\partial := \frac{\langle \nabla g_0, \nu\rangle}{\|\nabla g_0\|},$$

with ν *the outward unit normal of* $\mathcal{X}$, *and* no *cross term (there is no second perturbed surface to cross with). It vanishes when the margin meets* $\partial\mathcal{X}$ *orthogonally. The first derivative* (10) *is unaffected. Appendix C, item 6, verifies this display to ten digits. We flag it because the principal calibrated design of Section 6 violates (d): see Section 6.5, item (L3), which reports twelve such crossings.*

2 First-Order Theory

2.1 Pathwise derivative: two boundary integrals

For directions (u, v) (perturbations of (g_0, h_0) induced by perturbations of the four arm means), let $\gamma_t := (g_0 + tu,\ h_0 + tv)$.

Proposition 2.1 (First shape derivative)**.** *Under Assumptions 1.4–1.5, for directions* $(u, v) \in C^1(\mathcal{X})^2$,

$$D\Theta(\gamma_0)[u, v] := \frac{\mathrm{d}}{\mathrm{d}t}\Theta(\gamma_t)\Big|_{t=0} = \int_{\mathcal{M}_g} \frac{u(x)\, w(x)}{\|\nabla g_0\|}\, \mathrm{d}\mathcal{H}^{d-1}(x) + \int_{\mathcal{M}_h} \frac{v(x)\, w(x)}{\|\nabla h_0\|}\, \mathrm{d}\mathcal{H}^{d-1}(x). \quad (10)$$

Proof sketch. Apply the generalized Leibniz rule (Reynolds transport theorem; Delfour and Zolésio, 2001, Thm. 4.2) to the moving region $\Omega_t = \{g_t \geq 0\} \cap \{h_t \geq 0\}$ with fixed integrand w. The boundary $\partial\Omega_0$ decomposes, up to the $\mathcal{H}^{d-1}$-null corner, into the two faces $\mathcal{M}_g$ and $\mathcal{M}_h$; on each face the normal velocity is that of the corresponding level set, $-u/\|\nabla g_0\|$ and $-v/\|\nabla h_0\|$ respectively, and the coarea representation of the surface measure yields (10). □

Three features of (10) drive everything that follows.

(i) Irregularity. Each term is a $(d-1)$-dimensional Hausdorff integral: a bounded linear functional of (u, v) on C^1 but *not* on $L^2(f)$. Hence, exactly as in the single-threshold case, no L^2-Riesz representer (influence function) exists whenever w does not vanish on $\mathcal{M}_g \cup \mathcal{M}_h$, and Θ_0 is not $\sqrt{n}$-estimable. The welfare-type exception — boundary weights that vanish

on their own margins, as for total dynamic welfare V^* in the DTR section — is the regular special case.

(ii) Two bands. The derivative aggregates information about *two different contrasts* on *two different thin sets*. The variance of a plug-in estimator therefore has two band contributions, and, because $\hat{g}$ and $\hat{h}$ are estimated from the same observations, a within-unit covariance term appears whenever $\rho_{SY} \neq 0$.

(iii) The corner is invisible at first order. $\mathcal{C}$ has $\mathcal{H}^{d-1}$-measure zero and does not appear in (10); it enters only the second-order expansion (Section 3), which is precisely why it has been ignorable in first-order analyses and why it becomes relevant for debiasing.

2.2 Plug-in estimator, rate, and minimax optimality

Let $\hat{\mu}_S, \hat{\mu}_Y$ be stacked sieve least-squares estimators of the arm means on tensor-product B-spline bases of dimension K_n (per arm) and degree exceeding s — required for the approximation rate $\|b_g\|_\infty \lesssim K_n^{-s/d}$ used throughout, and raised further to degree 5 in Assumption 4.1(b) once the jackknife is introduced — let $(\hat{g}, \hat{h})$ be the implied contrast estimates, and let

$$\hat{\Theta} := \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{\hat{g}(X_i) \geq 0\}\, \mathbb{1}\{\hat{h}(X_i) \geq 0\}\, v_0(X_i)$$

be the (empirical-measure) plug-in estimator; in the known-density case the sample average is replaced by numerical integration against f.

Theorem 2.2 (Plug-in: rate, minimax optimality, and a centered CLT)**.** *Let Assumptions 1.4–1.6 hold together with the sieve regularity conditions of Chen et al. (2025, Thm. 3) applied to each of the four arm regressions, and write $K_n \asymp n^a$ and $\sigma_n^2 := \left\|v^*_{K_n}\right\|^2_{sd}$, with $v^*_{K_n}$ the sieve-Riesz representer of (10) defined in (12). Under Assumption 1.6(b), $\sigma_n^2 \asymp K_n^{1/d}$.*

- *(i) (*Rate.*) Let $a = d/(2s+1)$, $s \geq d$, and impose Assumption 1.8(b)–(c) in their $O_p(r_n^*)$ form. Then $\hat{\Theta} - \Theta_0 = O_p(r_n^*)$, $r_n^* = n^{-s/(2s+1)}$, with a constant depending on $\mathcal{P}_c^s$ only through $(c, \underline{c}, \underline{c}_T, \underline{w}, \underline{m}, \underline{p})$ and the sieve constants — hence uniformly over $\mathcal{P}_c^s$,* provided *the cited sieve regularity conditions and Assumption 1.8 hold uniformly over the class, which we assume but have not verified.*

- *(ii) (*Minimax.*) r_n^* is a minimax lower-bound rate over $\mathcal{P}_c^s$, so the plug-in rate in (i) is sharp.*

- *(iii) (*Centered CLT.*) Impose Assumption 1.8(b)–(d) in their $o_p(\mathrm{se}_n)$ form and*

$$\frac{d}{2s+1} \;<\; a \;<\; \min\left\{\frac{d}{2d+1},\, \frac{d}{3d-2}\right\}, \tag{11}$$

where the second edge is present only in the empirical-measure case (Remark 1.9); since $d/(2d{+}1) < d/(3d{-}2)$ iff $d < 3$, the quadratic edge binds for $d = 2$ and the equicontinuity edge for $d \geq 3$. The window is nonempty if and only if $s > d$ strictly *(known density, or $d = 2$), respectively $s > \max\{d, 3(d-1)/2\}$ (empirical measure, $d \geq 3$). Then $\sqrt{n}(\hat{\Theta} - \Theta_0)/\sigma_n \xrightarrow{d} \mathcal{N}(0,1)$, and the attained rate is $\mathrm{se}_n = n^{-(1-a/d)/2}$, slower than r_n^* but approaching it as $a \downarrow d/(2s+1)$.*

Proof sketch. Rate. The expansion (17) has a linear term that is a sum of two submanifold integrals of sieve LS errors; each is handled by the single-threshold argument (Chen and Gao, 2026, Thm. 8; Chen et al., 2025, Thm. 3) with the truncation indicators $\mathbb{1}\{h_0 > 0\}$, $\mathbb{1}\{g_0 > 0\}$ carried through as fixed bounded weights. Its stochastic part is $O_p(\sqrt{K_n^{1/d}/n}) = O_p(r_n^*)$ at $a = d/(2s+1)$ and its bias part is $O(K_n^{-s/d}) = O(r_n^*)$. The quadratic term's dominant piece is the margin stochastic diagonal, bounded by $K_n^{1+1/d}/n$, which Table 1 shows is $O(r_n^*)$ at $a = d/(2s+1)$ exactly when $s \geq d$. The third-order and equicontinuity remainders are supplied by Assumption 1.8(b)–(c); v3 omitted both from this part of the statement. *Minimax (nesting).* Consider the subclass of $\mathcal{P}_c^s$ in which $h_0 \geq \underline{\eta} > 0$ on the support, so that $\mathbb{1}\{h_0 \geq 0\} \equiv 1$ and Θ reduces to the single-threshold value functional $V(g_0)$. This subclass has $\mathcal{M}_h = \mathcal{C} = \emptyset$; it lies in $\mathcal{P}_c^s$ because Assumption 1.6(b) asks only that *one* margin be nondegenerate and (c) permits $\mathcal{C} = \emptyset$. (Had (b) been stated with "$\wedge$" the subclass would be empty and the argument would fail — see the remark following Assumption 1.6.) Every estimator of Θ_0 is then an estimator of $V(g_0)$, and the minimax lower bound of Chen and Gao (2026, Thm. 5) / Chen et al. (2025, Thm. 5) applies to that submodel. *Centered CLT.* Jointly asymptotically normal linear terms follow from the Cramér–Wold device applied to the stacked representer, with the Lyapunov condition Assumption 1.8(d) — which v3 invoked from Assumption 5.1(g), a hypothesis of a different theorem. For the limit to be *centered*, the linear bias, the quadratic diagonal and the equicontinuity term must all be $o(\mathrm{se}_n)$: $K_n^{-s/d} = o(\mathrm{se}_n)$ iff $a > d/(2s+1)$; $K_n^{1+1/d}/n = o(\mathrm{se}_n)$ iff $a < d/(2d+1)$; and (8) is $o(\mathrm{se}_n)$ iff $a < d/(3d-2)$. These are compatible iff $2s+1 > 2d+1$, respectively $2s+1 > 3d-2$. □

Remark 2.3 (Why the rate-optimal K_n does not give a centered limit)**.** *v1 and v2 asserted asymptotic normality at $K_n \asymp n^{d/(2s+1)}$ under $s \geq d$. This is not right, for two separate reasons, both visible in (11). First, at the rate-optimal K_n the sieve approximation bias contributes $D\Theta[b] = O(K_n^{-s/d}) = O(r_n^*)$ to the linear term, exactly the order of its standard deviation $\sqrt{K_n^{1/d}/n}$; the limit is normal but not centered. This is the usual undersmoothing requirement and it is not special to two thresholds. Second, at $s = d$ the margin quadratic diagonal $K_n^{1+1/d}/n$ is* exactly *$\asymp r_n^*$, hence $O(r_n^*)$ but not $o(r_n^*)$: enough for the rate statement* *(i), not for a centered limit. Strict $s > d$ is needed, with K_n chosen inside (11). In the revision note's language, "negligible under $s \geq d$" should read "no larger than the target rate under $s \geq d$"; negligibility needs the strict inequality. Remark 3.4 notes that this conclusion is conservative if the margin bound is not attained.*

2.3 Two-band sieve-Riesz variance

Let $\psi(x, w)$ denote the stacked per-arm sieve regressors of one outcome equation and $b(x) := (\psi^{(K_1)}(x)', -\psi^{(K_0)}(x)')'$ the contrast basis. The (infeasible) sieve-Riesz representer of (10) restricted to the sieve space is, for each outcome equation $o \in \{S, Y\}$,

$$v^*_{K_n,o}(x, w) := \psi(x, w)' R^{-1}_{K_n,o} \, D_o\Theta(\gamma_0)[\,\psi\,], \qquad R_{K_n,o} := \mathbb{E}[\psi(X_i, W_i)\psi(X_i, W_i)'], \tag{12}$$

where $D_o\Theta[\psi]$ stacks the coordinatewise boundary integrals of (10) over the relevant margin. Feasible versions replace population by empirical moments and the Hausdorff integrals by ε-band sample averages. Because an ε-band average of a function φ over $\{|\hat{g}| < \varepsilon\}$ converges by the coarea formula to $\int_{\{g_0=0\}} \varphi f_0 / \|\nabla g_0\| \, \mathrm{d}\mathcal{H}^{d-1}$ — against the *sampling* density f_0, not against $w = v_0 f$ — the band average must be reweighted by w/f_0:

$$\begin{aligned}
\widehat{D_g\Theta}[\,b\,] &= \frac{+1}{2\varepsilon_S}\,\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{|\hat{g}(X_i)| < \varepsilon_S,\ \hat{h}(X_i) > 0\}\,\frac{v_0(X_i) f(X_i)}{\hat{f}_0(X_i)}\,b(X_i), \\
\widehat{D_h\Theta}[\,b\,] &= \frac{-1}{2\varepsilon_Y}\,\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{|\hat{\tau}_Y(X_i)| < \varepsilon_Y,\ \hat{g}(X_i) \geq 0\}\,\frac{v_0(X_i) f(X_i)}{\hat{f}_0(X_i)}\,b(X_i).
\end{aligned} \tag{13}$$

In the empirical-measure case $f = f_0$ and the weight reduces to $v_0(X_i)$ — *no density estimate is required* — and in Example 1.1, where $v_0 \equiv 1$, it reduces to 1, which is the case implemented in Section 6. (v1 stated (13) without the weight, i.e. only for that case.) The known-density case $f \neq f_0$ does require $\hat{f}_0$, and then Proposition 2.4 needs the extra condition

$$\inf_{\mathcal{N}(\mathcal{M}_g \cup \mathcal{M}_h)} f_0 > 0, \qquad \left\|\hat{f}_0 - f_0\right\|_{\infty, \mathcal{N}(\mathcal{M}_g \cup \mathcal{M}_h)} = o_p(1), \tag{14}$$

on a neighborhood $\mathcal{N}(\cdot)$ of the margins, which in turn requires smoothness of f_0 that Assumption 1.4(d) does not impose (bounded gradient gives a uniform rate only for d small). We flag this rather than develop it: all results below are stated for the empirical-measure case, and the known-density case carries (14) as an additional maintained condition. The displays are instantiated for the harm-share convention of Example 1.1: for a generic pair (g, h) both bands carry a $+$ sign in (g, h) coordinates, and the $-$ sign of the second display is the Jacobian of $h_0 = -\tau_Y$ (raising τ_Y shrinks the harmed region).

The variance estimator sums the *per-unit* influence contributions of the two outcome equations *before* squaring, and adds the empirical-measure term:

$$\widehat{\mathrm{Var}}(\hat{\Theta}) = \frac{\hat{\sigma}_n^2}{n} + \frac{\hat{\varsigma}^2}{n}, \quad \hat{\sigma}_n^2 := \frac{1}{n}\sum_{i=1}^{n}\Big(\widehat{v_S^*}(X_i, W_i)\,\hat{\epsilon}_{S,i} + \widehat{v_Y^*}(X_i, W_i)\,\hat{\epsilon}_{Y,i}\Big)^2, \tag{15}$$

where $\hat{\varsigma}^2 := n^{-1}\sum_i \big(\mathbb{1}\{\hat{g}_i \geq 0\}\mathbb{1}\{\hat{h}_i \geq 0\}v_0(X_i) - \hat{\Theta}\big)^2$ reduces to $\hat{\Theta}(1-\hat{\Theta})$ when $v_0 \equiv 1$. Summing the two influence contributions before squaring means the within-unit covariance of the short- and long-run residuals ($\rho_{SY} \neq 0$) is captured automatically. Here the stacked standard-deviation norm is defined on the *summed* per-unit influence,

$$\|(v_S, v_Y)\|_{sd}^2 := \mathbb{E}\big[\{v_S(X_i, W_i)\epsilon_{S,i} + v_Y(X_i, W_i)\epsilon_{Y,i}\}^2\big], \tag{16}$$

not as a scalar $\mathbb{E}[v^2\sigma^2]$: the within-unit correlation ρ_{SY} enters σ_n^2 through the cross term of (16) and would otherwise be lost. $v_{K_n}^*$ denotes the stacked (S, Y) representer. Adding the two components of (15) ignores their covariance — the same X_i enter the empirical measure and the first-stage residuals — which is of smaller order than either and which we do not estimate. Because $\sigma_n^2 \asymp K_n^{1/d} \to \infty$ while $\hat{\varsigma}^2 = O(1)$, the second term of (15) is asymptotically negligible; it is 6–12% of the reported standard error at the sample sizes of Section 6, and omitting it produces a spurious SE/SD ratio of 0.87–0.95.

Proposition 2.4 (Ratio consistency)**.** *Let Assumptions 1.4–1.6 hold in the empirical-measure case, and suppose: (a) $\varepsilon_n \downarrow 0$ with $\varepsilon_n^{-1}(\|\hat{g} - g_0\|_\infty + \big\|\hat{h} - h_0\big\|_\infty) = o_p(1)$ and $n\varepsilon_n/(K_n \log K_n) \to \infty$; (b) the empirical Gram matrix satisfies $\big\|\hat{R}_{K_n} R_{K_n}^{-1} - I\big\|_{\mathrm{op}} = o_p(1)$, which for tensor B-splines holds when $K_n \log K_n / n \to 0$; (c) the feasible band vectors converge in the representer norm, $\big\|\hat{v}_{K_n}^* - v_{K_n}^*\big\|_{sd} = o_p(\sigma_n)$; (c′) the four* arm *residuals used in (15) are consistent, $n^{-1}\sum_i\{\hat{\epsilon}_{o,i} - \epsilon_{o,i}\}^2 = o_p(1)$ for $o \in \{S, Y\}$ — consistency of the* contrasts *$\hat{g}, \hat{h}$ does not imply it, since a common additive error in $\hat{\mu}_o(\cdot, 1)$ and $\hat{\mu}_o(\cdot, 0)$ cancels from the contrast but not from the residuals; (d) (*known-density case only*) (14) holds. Then $\hat{\sigma}_n^2/\sigma_n^2 \xrightarrow{p} 1$, and whenever $\sqrt{n}(\hat{\Theta} - \Theta_0)/\sigma_n$ has a centered normal limit (Theorem 2.2(iii)) so does the studentized statistic $\sqrt{n}(\hat{\Theta} - \Theta_0)/\hat{\sigma}_n$.*

Conditions (b) and (c) are new relative to v1–v2, which imposed only (a). They are not decoration: (13) estimates a K_n-vector from the $\asymp n\varepsilon_n$ observations that fall in a band and then inverts a $K_n \times K_n$ Gram matrix against it, so an empirical-process argument with the usual logarithmic factor is required, and *consistency of $\hat{v}^*$ in the standard-deviation norm* is what makes $\hat{\sigma}_n^2$ estimate σ_n^2 — a product-rate condition of the kind used in Theorem 5.3

does not imply it. We have not verified (b)–(c) from primitive conditions; see Appendix B, §B.4.

Ratio consistency is the relevant notion because $\sigma_n^2 \to \infty$ in the irregular regime. The proof adapts the single-threshold argument to each band, using Evans and Gariepy (2015, Thm. 3.13(iii)) for the ε-band-to-Hausdorff convergence on each truncated margin. The single-threshold statement requires only $n\varepsilon_n \to \infty$; we impose the stronger $n\varepsilon_n/K_n \to \infty$ because (13) estimates a K_n-vector from the $\asymp n\varepsilon_n$ observations that fall in the band, and the feasible representer inverts a $K_n \times K_n$ Gram matrix against it. This is a sufficient-side strengthening, not an imported result. The requirement $\varepsilon_n \to 0$ is essential: a bandwidth that stabilizes at a fixed positive fraction of $\mathrm{sd}(\hat{\tau})$ estimates a smeared functional, not $D\Theta$, and Proposition 2.4 does not apply to it. See Section 6.5, item (L5).

3 Second-Order Theory: the Corner

3.1 The second pathwise derivative

The plug-in expansion

$$\hat{\Theta} - \Theta_0 = D\Theta(\gamma_0)[\hat{\gamma} - \gamma_0] + \tfrac{1}{2} D^2\Theta(\gamma_0)[\hat{\gamma} - \gamma_0, \hat{\gamma} - \gamma_0] + R_3, \tag{17}$$

with R_3 a third-order remainder, is what limits the smoothness requirement of Theorem 2.2. The new object is $D^2\Theta$.

Theorem 3.1 (Second derivative: two margins and a corner)**.** *Under Assumptions 1.4–1.5, with $n_g := \nabla g_0 / \|\nabla g_0\|$, $H_g := \mathrm{div}(n_g)$ the mean curvature of $\{g_0 = 0\}$ (and symmetrically for h), Θ is twice pathwise differentiable with*

$$D^2\Theta(\gamma_0)[(u,v),(u,v)] = Q_g[u,u] + Q_h[v,v] + 2\int_{\mathcal{C}} \frac{u(x)\,v(x)\,w(x)}{\|\nabla g_0\|\;\|\nabla h_0\|\;|\sin\omega(x)|}\,\mathrm{d}\mathcal{H}^{d-2}(x), \tag{18}$$

where the margin quadratic forms are

$$\begin{aligned} Q_g[u,u] = -\int_{\mathcal{M}_g} \left[\frac{u\,(n_g\cdot\nabla u)\,w}{\|\nabla g_0\|^2} + \frac{u}{\|\nabla g_0\|}\left\{ n_g\cdot\nabla\Big(\frac{u\,w}{\|\nabla g_0\|}\Big) + \frac{u\,w}{\|\nabla g_0\|}\,H_g \right\}\right] \mathrm{d}\mathcal{H}^{d-1} \\ - \int_{\mathcal{C}} \frac{u^2(x)\,w(x)\,\cos\omega(x)}{\|\nabla g_0\|^2\;|\sin\omega(x)|}\,\mathrm{d}\mathcal{H}^{d-2}(x), \end{aligned} \tag{19}$$

and Q_h is the mirror image (swap $g \leftrightarrow h$, $u \leftrightarrow v$, and restrict to $\mathcal{M}_h$).

Proof sketch; full derivation in Appendix A. Differentiate each term of (10) along γ_t. Three effects arise for the $\mathcal{M}_g$-term: (i) the integrand $u\,w/\,\|\nabla g_t\|$ changes (the $n_g{\cdot}\nabla u$ term); (ii) the level set $\{g_t = 0\}$ flows with normal velocity $-u/\,\|\nabla g_0\|$, producing the tangential-gradient and curvature terms of the moving-level-set lemma of Chen and Gao (2026, Lem. PathD–Level) applied with weight $u\,w/\,\|\nabla g_0\|$; (iii) the *truncation* $\{h_t > 0\}$ of $\mathcal{M}_g$ moves. Effect (iii) has two parts: the direct motion of the cut in the v-direction, an edge integral over $\mathcal{C}$ with within-manifold coarea factor $\|P_T \nabla h_0\| = \|\nabla h_0\|\,|\sin\omega|$ (this is the cross term, and it appears once from each margin, giving the factor 2 in (18)); and the induced motion of the edge caused by the flow of the manifold itself across the fixed cut — pulling the cut condition back through the normal flow perturbs it by $-tu\,\|\nabla h_0\|\cos\omega/\,\|\nabla g_0\|$, which yields the diagonal corner term in (19). Transversality (Assumption 1.5(b)) keeps all corner factors bounded. ☐

When $d = 2$ the corner is a finite set of points and $\mathcal{H}^0$ is counting measure: the cross term is $2\sum_{x\in\mathcal{C}} u(x)v(x)w(x)/(\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|)(x)$.

Remark 3.2 (Bilinear extension)**.** *Theorem 3.1 is stated on the diagonal, $D^2\Theta[\eta,\eta]$, but is used below as a symmetric bilinear form on pairs of* distinct *directions — $D^2\Theta[u^A, u^B]$ in Theorem 4.6, $D^2\Theta[e_A, e_B]$ in Theorem 5.8, $C[\Delta_g, \Delta_h]$ throughout. The extension is by polarization,*

$$D^2\Theta[\eta_1,\eta_2] := \tfrac{1}{4}\bigl\{D^2\Theta[\eta_1+\eta_2,\eta_1+\eta_2] - D^2\Theta[\eta_1-\eta_2,\eta_1-\eta_2]\bigr\}, \tag{20}$$

valid on the direction space $C^1(\mathcal{X})^2$, on which the right-hand side of (18) is a bounded symmetric quadratic form under Assumptions 1.4–1.5. Explicitly, writing $\eta_k = (u_k, v_k)$, the polarized form is $Q_g[u_1,u_2]+Q_h[v_1,v_2]+C[u_1,v_2]+C[u_2,v_1]$, with Q_g and C polarized in the same way. All bilinear manipulations below — in particular the split-sample identity (31) — are manipulations of (20).

Remark 3.3 (Numerical verification)**.** *Theorem 3.1 is verified to ten significant digits in Appendix C on closed-form $d = 2$ designs that between them activate every term of (18)–(19): a radially weighted disk (the transport and curvature terms of Q_g); a disk with a non-constant direction u (the $u\,n_g{\cdot}\nabla u$ term); a disk cut by an oblique chord and the intersection of two disks (both corner diagonals and the corner cross term, at unequal u, v, at one-sided perturbations, and at both signs of $\cos\omega$). The same appendix verifies the polarization identity (37) and the support-edge term of Remark 1.10. This is the part of the analysis that is checked rather than asserted.*

3.2 Orders of the second-order terms

Write $\Delta_g := \hat{g} - g_0$, $\Delta_h := \hat{h} - h_0$ for sieve LS first stages with K_n basis functions per arm. Decompose each error into an approximation bias and a linear stochastic part,

$$\Delta_g = b_g + u_g, \qquad u_g(x) := \frac{1}{n}\sum_i s_i(x)\,\epsilon_{S,i}, \qquad s_i(x) = b(x)'G^{-1}b(X_i), \tag{21}$$

with $\|b_g\|_\infty \lesssim K_n^{-s/d}$, $\|\nabla b_g\|_\infty \lesssim K_n^{-(s-1)/d}$, $G := \mathbb{E}[b(X_i)b(X_i)']$, and similarly for Δ_h with residuals $\epsilon_{Y,i}$. Note that the sieve influence weights s_i are *common* to the two regressions, because the same design matrix and the same basis are used; this is what generates the within-unit coupling below.

Two scaling facts for tensor-product B-splines with mesh $\varkappa = K_n^{-1/d}$ drive the whole accounting:

$$\mathcal{V}(x) := b(x)'G^{-1}b(x) \asymp K_n, \qquad \left\|G^{-1/2}\nabla b(x)\right\| \lesssim K_n^{1/2+1/d}, \tag{22}$$

(each derivative of a basis function costs a factor $\varkappa^{-1} = K_n^{1/d}$), whence $|b(x)'G^{-1}\partial_j b(x)| \lesssim K_n^{1+1/d}$ by Cauchy–Schwarz. Only the first display is two-sided; the second bounds the norm of a *signed* vector, and v2's "$b'G^{-1}\nabla b \asymp K_n^{1+1/d}$" must be read that way. The distinction is not cosmetic: Remark 3.4 shows that the integrated version of the signed quantity can be an order smaller.

The estimator runs *four separate arm regressions* on the common basis ψ, so the exact second moments are arm-wise sums of sandwiches, not a single one. Write $\pi_1(x) := p_0(x)$, $\pi_0 := 1 - \pi_1$, and

$$R_w := \mathbb{E}\big[\pi_w(X)\psi(X)\psi(X)'\big], \qquad \Omega^w_{oo'} := \mathbb{E}\big[\pi_w(X)\,\varsigma_{oo'}(X,w)\,\psi(X)\psi(X)'\big],$$

for $o, o' \in \{S, Y\}$, with $\varsigma_{SS} = \sigma_S^2$, $\varsigma_{YY} = \sigma_Y^2$ and $\varsigma_{SY} = \rho_{SY}\sigma_S\sigma_Y$. Since $\mathbb{1}\{W_i = 1\}\mathbb{1}\{W_i = 0\} = 0$, cross-arm covariances vanish and, with $g_0 = \tau_S$ and $h_0 = -\tau_Y$ as in Example 1.1,

$$\mathcal{V}_{gh,n}(x) := n\,\mathbb{E}[u_g(x)u_h(x)] = -\sum_{w\in\{0,1\}} \psi(x)'R_w^{-1}\Omega^w_{SY}R_w^{-1}\psi(x), \tag{23}$$

and $\mathcal{V}_{gg,n}, \mathcal{V}_{hh,n}$ likewise with Ω^w_{SS}, Ω^w_{YY} and no leading minus. *The minus sign is the Jacobian of* $h_0 = -\tau_Y$: the estimation error of h is the negative of that of τ_Y, so a positive ρ_{SY} produces a *negative* $\mathcal{V}_{gh,n}$. v3 omitted this.

Under uniform overlap and Assumption 1.4(c), $\mathcal{V}_{gg,n}(x) \asymp \mathcal{V}_{hh,n}(x) \asymp K_n$ and $|\mathcal{V}_{gh,n}(x)| \lesssim K_n$ uniformly. If in addition the ς's and π_w are Lipschitz, local support of ψ gives the *additive* approximation

$$\mathcal{V}_{gg,n}(x) = \varrho_{gg}(x)\,\mathcal{V}(x) + O(\varkappa K_n), \qquad \left|\mathbb{E}[u_g(x)\,(n_g\cdot\nabla u_g)(x)]\right| \lesssim \frac{K_n^{1+1/d}}{n},$$
$$\mathcal{V}_{gh,n}(x) = -\varrho_{gh}(x)\,\mathcal{V}(x) + O(\varkappa K_n), \tag{24}$$

where $\mathcal{V}(x) := \psi(x)'R^{-1}\psi(x)$ with $R := \mathbb{E}[\psi\psi']$, and

$$\varrho_{gg}(x) := \sum_{w\in\{0,1\}} \frac{\sigma_S^2(x,w)}{\pi_w(x)}, \qquad \varrho_{gh}(x) := \sum_{w\in\{0,1\}} \frac{\rho_{SY}(x,w)\,\sigma_S(x,w)\,\sigma_Y(x,w)}{\pi_w(x)}, \tag{25}$$

ϱ_{hh} symmetrically. The error term in (24) is *additive*, $O(\varkappa K_n)$, not relative: a relative $1 + O(\varkappa)$ statement is vacuous — indeed false — near points where the signed ϱ_{gh} vanishes, which is exactly where the sign of the corner contribution turns over. Under conditional homoskedasticity and constant propensity the error is identically zero.

Three corrections are recorded here. v1 wrote the covariance display with an undefined normalization and "$\asymp$" applied to a signed quantity; v2 replaced it by an exact identity that holds only under homoskedasticity; v3 wrote a single-sandwich form appropriate to one pooled contrast regression rather than the four arm regressions actually used, with a relative error term and without the $h_0 = -\tau_Y$ sign. The consequences: the sign of $\mathcal{V}_{gh,n}(x)$ is *not* that of $\rho_{SY}(x,w)$ — it is the opposite, under the harm-share convention — and a nonzero conditional correlation at a corner does not by itself guarantee a nonzero *integrated* corner bias, which is why Assumption 1.6(c) is imposed on the normalized sequence rather than derived from $\rho_{SY} \neq 0$. The stochastic orders below survive as upper bounds; the constants do not.

Taking expectations term by term in (18) gives the *upper bounds* in Table 1. The essential point — and the correction to v1 — is that the margin forms Q_g, Q_h contain a factor ∇u while the corner forms do not. *The margin diagonals are therefore bounded by $K_n^{1/d}$ times the corner diagonals*, not by the same quantity. Every entry of Table 1 is an $O(\cdot)$, and this is all the theorems below use; Remark 3.4 discusses which entries are actually attained.

Three consequences follow, and they replace the corresponding discussion of v1.

(a) The plug-in's smoothness requirement comes from the margin diagonal. $K_n^{1+1/d}/n = n^{-(2s-d)/(2s+1)}$ is $O(r_n^*)$ if and only if $s \geq d$ — precisely the condition in Theorem 2.2. In v1 this requirement was attributed to K_n/n, which would have given $s \geq d-1$, and the one-derivative discrepancy was left unexplained. With the bound of Table 1 the condition $s \geq d$ is exactly what is needed for the bound to be negligible. We do *not* claim it is necessary; see Remark 3.4.

Table 1: Upper bounds on the second-order terms and the smoothness needed for each to be $O(r_n^*)$, $r_n^* = n^{-s/(2s+1)}$, at $K_n \asymp n^{d/(2s+1)}$. "Diagonal" terms enter the estimator error linearly and must be compared with r_n^* itself; "off-diagonal" (cross-half / $i \neq j$) terms are mean-zero degenerate U-statistics. Derivations of the off-diagonal orders are in Appendix B. Entries are bounds, not exact orders: see Remark 3.4.

Component	Term	Bound	bound is $O(r_n^*)$ iff
Stochastic diagonals (removed by SS/LOO debiasing)			
Margin	$Q_g[u_g, u_g]$, $Q_h[u_h, u_h]$	$K_n^{1+1/d}/n$	$s \geq d$
Corner	all three corner pieces	K_n/n	$s \geq d-1$
Approximation-bias diagonals (removed by undersmoothing)			
Margin	$Q_g[b_g, b_g]$	$K_n^{-(2s-1)/d}$	$s > 1$
Corner	$C[b_g, b_h]$ etc.	$K_n^{-2s/d} = r_n^{*2}$	always
Bias $\times$ stochastic cross terms (removed by neither)			
Margin	$Q_g[b_g, u_g]$	$K_n^{(1-s)/d}\sqrt{K_n/n}$	$s \geq (d+1)/2$
Corner	$C[b_g, u_h]$	$K_n^{-s/d}\sqrt{K_n/n}$	$s \geq (d-1)/2$
Cross-half off-diagonals (mean zero; degenerate)			
Margin	gradient part	$n^{-1}K_n^{(d+3)/(2d)}$	$s \geq (d+1)/2$
Corner	—	$n^{-1}K_n^{(d+2)/(2d)}$	$s \geq d/2$
Margin	curvature part	$n^{-1}K_n^{(d+1)/(2d)}$	$s \geq (d-1)/2$

(b) The corner is no larger than the margins and is still rate-relevant on a nonempty window. $K_n/n = n^{-(2s+1-d)/(2s+1)}$ is $O(r_n^*)$ if and only if $s \geq d-1$. Since split-sample debiasing (Section 4) extends the guarantee down to $s > (d+1)/2$, the corner diagonal is rate-relevant — i.e. larger than r_n^* inside the regime where the theory is supposed to deliver — exactly on

$$\tfrac{d+1}{2} < s \leq d-1, \qquad \text{nonempty iff } d \geq 4. \tag{26}$$

For $d \in \{2,3\}$ the window is empty, so at the $d = 2$ of our designs corner-awareness is a *finite-sample constants* question, not an asymptotic rate question. This is a weaker claim than v1's abstract made, and it is the correct one.

(c) The corner coefficient has three pieces, not one. Collecting the corner terms of (18)–(19) and using (24), the corner's contribution to $\mathbb{E}[\frac{1}{2}D^2\Theta[\Delta,\Delta]]$ is

$$\mathcal{B}_{\mathcal{C},n} = \int_{\mathcal{C}} \frac{w(x)}{2\,|\sin\omega(x)|}\left[\frac{2V_{gh}(x)}{\|\nabla g_0\|\,\|\nabla h_0\|} - \cos\omega(x)\left(\frac{V_g(x)}{\|\nabla g_0\|^2} + \frac{V_h(x)}{\|\nabla h_0\|^2}\right)\right]\mathrm{d}\mathcal{H}^{d-2}(x), \tag{27}$$

with cross-term component

$$\mathcal{B}^{\times}_{\mathcal{C},n} := \int_{\mathcal{C}} \frac{V_{gh}(x)\,w(x)}{\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega(x)|}\,\mathrm{d}\mathcal{H}^{d-2}(x), \tag{28}$$

where $V_g := \mathbb{E}[\Delta_g^2] = \mathcal{V}_{gg,n}/n + b_g^2$, V_h likewise, and $V_{gh} := \mathbb{E}[\Delta_g\Delta_h] = \mathcal{V}_{gh,n}/n + b_g b_h$, all at x. All three are $O(K_n/n)$ up to their bias parts, so $\mathcal{B}_{\mathcal{C},n} = O(K_n/n)$ — and *only* O. The bracket is a *signed* difference, so its three components may offset; contributions from different corner points may offset; and the weights $\mathcal{V}_{\bullet,n}(x)/K_n$ are n-dependent and vary across corner points. Matching lower bounds are exactly (5), which is why those are imposed on the normalized sequences $\frac{n}{K_n}\mathcal{B}_{\mathcal{C},n}$ and $\frac{n}{K_n}\mathcal{B}^{\times}_{\mathcal{C},n}$ rather than on population coefficients. v2 wrote "$\mathcal{B}_{\mathcal{C}} \asymp K_n/n$"; that is a two-sided claim the formula does not support on its own, and v3's population-level noncancellation condition did not repair it. Two further readings matter. First, the sign of the corner bias is not determined by $\operatorname{sign}(\rho_{SY})$ alone — nor, by (23), is $\operatorname{sign}(V_{gh}(x))$ determined by $\operatorname{sign}(\rho_{SY}(x,w))$. Second, the corner contribution to the *plug-in* bias does not vanish when $\rho_{SY} = 0$: the two $\cos\omega$ terms survive, and they vanish only when the margins meet orthogonally. In the notation of (27), v1 tracked only the first term of the bracket. Note carefully, however, that the two $\cos\omega$ terms belong to Q_g and Q_h and are therefore removed by margin-by-margin debiasing; only the V_{gh} term survives it, and *that* term does vanish at $\rho_{SY} = 0$. Section 4 makes this precise (Proposition 4.10), and the two statements should not be conflated: (27) is the corner's contribution to the plug-in's bias, not the part that margins-only debiasing leaves behind.

Remark 3.4 (Are these bounds attained? A caution about mesh resonance)**.** *Table 1 reports bounds, and for the margin diagonal the bound need not be tight. Under conditional homoskedasticity the leading margin diagonal is, exactly,*

$$\mathbb{E}\big[Q_g[u_g,u_g]\big]_{grad} = -\frac{\sigma^2}{n}\int_{\mathcal{M}_g}\frac{w}{\|\nabla g_0\|^2}\,n_g\cdot\nabla\mathcal{V}\,\mathrm{d}\mathcal{H}^{d-1}, \qquad \mathcal{V}(x) := b(x)'G^{-1}b(x), \tag{29}$$

because $\mathbb{E}[u_g\,\partial_n u_g] = (\sigma^2/2n)\,\partial_n\mathcal{V}$*. Now* $\mathcal{V}\asymp K_n$ oscillates *at the mesh scale* $\varkappa = K_n^{-1/d}$*, so* $\|\nabla\mathcal{V}\|_\infty \asymp K_n^{1+1/d}$ *pointwise — but (29) integrates it against a smooth weight over a* $(d-1)$*-dimensional surface, and an oscillating integrand can cancel. Whether it does depends on the arithmetic relationship between the margin's normal direction and the tensor mesh. Writing* $\mathcal{V}$ *as a Fourier series in the mesh period, the surface integral retains a non-decaying contribution only from* resonant *frequency pairs, which requires the margin's slope to be rational with a small denominator. Numerically (Appendix C, item 5), with periodic cubic tensor B-splines in* $d=2$*: a mesh-aligned margin gives* $\asymp K_n^{1+1/d}$*, the exact diagonal* $\alpha=1$ *gives the same order with a smaller constant, and generic irrational slopes* ($1/\pi$, $\sqrt{2}$, $\sqrt{5}-1$) *give* $\asymp K_n$ *— a hundredfold gap at* $K_n^{1/2} = 128$.

Three implications. (i) $s\ge d$ *in Theorem 2.2(i) is sufficient but we do not claim it necessary. If the margin diagonal is in fact* K_n/n*, the rate condition relaxes to* $s\ge d-1$ *and the centered-CLT window (11) widens to* $d/(2s+1) < a < d/(2d-1)$*, nonempty iff* $s > d-1$ *— one derivative cheaper throughout. Nothing below relies on this; we record it because it is the difference between the plug-in needing* $s>d$ *and* $s>d-1$*.* (ii) *This particular mechanism does not reach the corner:* V_g, V_h, V_{gh} *in (27) involve* $\mathcal{V}$ undifferentiated *and are evaluated pointwise on a* $(d-2)$*-set, so there is no oscillating derivative to average away, and each of the three is* $\asymp K_n/n$ *individually. This does* not *make* $\mathcal{B}_{\mathcal{C}}$ *itself two-sided: as noted above, the corner coefficient can still cancel through the signed bracket or across corner points, which is a geometric and covariance phenomenon rather than a mesh phenomenon, and which Assumption 1.6(c) excludes by assumption. v2's claim that "the corner bound is the robust one" should be read narrowly: robust to* mesh resonance*, not to cancellation in general.* (iii) *Consequently the safe statement of the margin-versus-corner comparison is an inequality: the corner is never more than* $K_n^{1/d}$ *smaller than the margins, and on a generic oblique geometry the two are plausibly of the same order — which is where v1's instinct, though not its argument, pointed. Two caveats on the mechanism itself: the identity* $\mathbb{E}[u\,\partial_n u] = (\sigma^2/2n)\partial_n\mathcal{V}$ *requires homoskedasticity (in general the object is* $n^{-1}b'G^{-1}\Omega_{gg}G^{-1}\nabla b$ *by (23), not an exact gradient), and the evidence is a* $d=2$ *probe with straight margins. Establishing the exact order of (29) for curved margins and general* d *is open; we flag it because it is the one place where the accounting above could be conservative rather than wrong.*

A margin-by-margin application of the single-threshold correction of Chen and Gao (2026) removes the margin diagonals *and* the two $\cos\omega$ corner diagonals, but leaves the corner cross term $\mathcal{B}_{\mathcal{C}}^{\times}$ in place; Proposition 4.10 below records the implication.

4 Quadratic Debiasing

4.1 Split-sample estimator and a tuning-free representation

Randomly split the sample into halves A and B (stratified by W), run the stacked sieve regressions on each half, and let $\hat\gamma^A = (\hat g^A, \hat h^A)$, $\hat\gamma^B$ denote the two estimated pairs, $\bar\gamma := (\hat\gamma^A + \hat\gamma^B)/2$, and $\Delta := \hat\gamma^A - \hat\gamma^B$. The split-sample (SS) debiased estimator is

$$\hat\Theta_{SS} := \Theta(\bar\gamma) - \tfrac{1}{8}\, D^2\Theta(\bar\gamma)[\Delta, \Delta]. \tag{30}$$

The construction works because the approximation bias is common to the two halves and cancels from Δ: writing $\hat\gamma^j = \gamma_0 + b + u^j$ with a common deterministic b and independent mean-zero u^A, u^B,

$$\bar\Delta := \bar\gamma - \gamma_0 = b + \bar u, \quad \bar u = \tfrac{1}{2}(u^A + u^B), \qquad \Delta = u^A - u^B, \tag{31}$$

and by bilinearity $D^2\Theta[\bar u, \bar u] - \frac{1}{4}D^2\Theta[\Delta, \Delta] = D^2\Theta[u^A, u^B]$, a mean-zero cross-half term. Thus (30) removes the *stochastic* diagonals — margins and corner alike — and nothing else. It does not touch the bias diagonals or the bias $\times$ stochastic cross terms; those are the third and fourth blocks of Table 1 and are controlled by smoothness, not by debiasing.

Assumption 4.1 (Path regularity)**.** *(a) (*Weight.*) $w = v_0 f \in C^3$ in a neighborhood of $\{g_0 = 0\} \cup \{h_0 = 0\}$. This is a restriction on the user-chosen v_0 and on the target density f; it is* not *a restriction on $\mu_{S,0}, \mu_{Y,0}$ and therefore does not interact with s. (b) (*Sieve order.*) The first stages take values in a space $\mathcal{S} \subset C^4(\mathcal{X})$ — for tensor B-splines, degree at least 5 (quintic or higher), since a degree-p B-spline is only C^{p-1} at its knots. The order must in any case exceed s for the approximation rate $\|b_g\|_\infty \lesssim K_n^{-s/d}$ used throughout, so the operative requirement is degree $\geq \max\{5, \lceil s \rceil\}$. (c) (*Uniform geometry along the path.*) With probability tending to one, every γ on the segment $\{\bar\gamma + t\Delta : |t| \leq \frac{1}{2}\}$ satisfies Assumption 1.5(a)–(d) with constants $\underline{c}/2, \underline{c}_T/2$. (d) (*Fourth-derivative bound.*) On that segment $t \mapsto \Theta(\bar\gamma + t\Delta)$ is four times continuously differentiable with $\sup_{|t|\leq 1/2} \left| \frac{\mathrm{d}^4}{\mathrm{d}t^4}\Theta(\bar\gamma + t\Delta) \right| \lesssim \|\Delta\|_{C^1}^4$.*

Three comments. First, v2 stated (a)–(d) as a single condition requiring a C^3 *neighborhood of* γ_0, which implicitly forces $\gamma_0 \in C^3$ and hence $s \geq 3$ — flatly inconsistent with a theorem claiming performance over Λ^s_c with, say, $1.5 < s < 3$ at $d = 2$. The smoothness a fourth-order finite difference needs is smoothness of the *path along which* Θ *is differenced*, i.e. of the *fits* and of w, not of the truth: the level sets being deformed are those of $\bar\gamma + t\Delta$. Splitting the condition this way removes the inconsistency. Second, (b) is a real implementation constraint that v2 overlooked: cubic B-splines are only C^2 at their knots, so the third and fourth derivatives that (d) presumes do not exist for the maintained implementation of Section 6. The order needed follows from the derivative count in Appendix A: $D^k\Theta$ involves k derivatives of the level-set function and $k-1$ of the weight and of the perturbation direction — D^2 already carries H_g (two derivatives of g), ∇w, and ∇u. Hence D^4 needs C^4 level-set functions and C^3 weight and direction. Since $\bar\gamma$ and Δ inhabit the same sieve space, that space must be C^4: degree 5, not degree 4 as an earlier draft of this revision had it. The calibrated Monte Carlo uses cubics and therefore satisfies neither. Third, (c) holds for sieve first stages on an event of probability tending to one whenever $\|\hat\gamma^j - \gamma_0\|_{C^1} = o_p(1)$, i.e. $K_n^{1/2+1/d} n^{-1/2} + K_n^{-(s-1)/d} = o(1)$, which needs $s > 1$ and is implied by (41).

Lemma 4.2 (Half-sample jackknife representation)**.** *Under Assumption 4.1, approximating* $D^2\Theta(\bar\gamma)[\Delta,\Delta]$ *by the symmetric second difference at step* $\delta = \frac{1}{2}$ *— i.e. evaluating* Θ *at* $\bar\gamma \pm \frac{1}{2}\Delta = \hat\gamma^A, \hat\gamma^B$ *— gives the closed form*

$$\hat\Theta^{\text{jack}}_{SS} = 2\,\Theta(\bar\gamma) - \tfrac{1}{2}\big(\Theta(\hat\gamma^A) + \Theta(\hat\gamma^B)\big). \tag{32}$$

Writing $\phi(t) := \Theta(\bar\gamma + t\Delta)$*, the approximation error is exactly*

$$\hat\Theta^{\text{jack}}_{SS} - \hat\Theta_{SS} = -\tfrac{1}{384}\,\phi^{(4)}(\xi) \quad \textit{for some } |\xi| \le \tfrac{1}{2}.$$

Under the fourth-derivative bound of Assumption 4.1, $\phi^{(4)}(\xi) = O_p(\|\Delta\|^4_{C^1}) = O_p(K_n^{2+4/d} n^{-2})$*, which is* $o(r^*_n)$ *at* $K_n \asymp n^{d/(2s+1)}$ *if and only if*

$$s > \tfrac{2(d+1)}{3}. \tag{33}$$

Moreover (32) is exact whenever $t \mapsto \Theta(\bar\gamma + t\Delta)$ *is a quadratic polynomial.*

Proof. $\phi(\frac{1}{2}) + \phi(-\frac{1}{2}) - 2\phi(0) = \frac{1}{4}\phi''(0) + \frac{1}{192}\phi^{(4)}(\xi)$ by Taylor's theorem with Lagrange remainder, valid because $\phi \in C^4[-\frac{1}{2}, \frac{1}{2}]$ under Assumption 4.1. Substituting $D^2\Theta(\bar\gamma)[\Delta,\Delta] = \phi''(0)$ into (30) and rearranging gives (32) with the stated error. The rate statement substitutes $\|\Delta\|_{C^1} \asymp K_n^{1/2+1/d} n^{-1/2}$ and $K_n \asymp n^{d/(2s+1)}$, giving exponent $(2d + 2 - 3s)/(2s + 1)$ relative to r^*_n. ∎

Remark 4.3 (An open gap in the jackknife implementation)**.** *Condition* (33) *is* strictly stronger *than the $s > (d+1)/2$ of Theorem 4.6: $2(d+1)/3 > (d+1)/2$ for every d, e.g. $s > 2$ against $s > 1.5$ at $d = 2$. So with the bound currently available, the tuning-free implementation* (32) *is not proved to inherit the weak-smoothness guarantee of the estimator* (30) *that it approximates; only $\hat{\Theta}_{SS}$ itself, computed from the closed-form $D^2\Theta$ of Theorem 3.1, is covered on the full range $s > (d+1)/2$. The gap may well be an artifact of the crude bound $\phi^{(4)} \lesssim \|\Delta\|_{C^1}^4$ assumed in Assumption 4.1, which charges a gradient factor $K_n^{1/d}$ to each of the four directions. The second derivative* (18)–(19) *charges only* one *gradient across its two directions. Substituting $\|\Delta\|_\infty \asymp \sqrt{K_n/n}$ for each direction that carries no gradient gives the ladder*

$$\begin{aligned}
\phi^{(4)} &\lesssim \|\Delta\|_{C^1}^4 && \text{(4 gradients)} \implies s > \tfrac{2(d+1)}{3},\\
\phi^{(4)} &\lesssim \|\Delta\|_\infty \|\Delta\|_{C^1}^3 && \text{(3 gradients)} \implies s > \tfrac{2d+1}{3},\\
\phi^{(4)} &\lesssim \|\Delta\|_\infty^2 \|\Delta\|_{C^1}^2 && \text{(2 gradients)} \implies s > \tfrac{2d}{3},
\end{aligned}$$

the last of which is weaker *than $s > (d+1)/2$ for every $d \geq 2$, so the gap would close entirely. Which of the three holds is decided by the third and fourth shape derivatives of Θ, which we have not computed; v1 asserted the $o_p(r_n^*)$ conclusion without either the assumption or the calculation. Until then the honest statement is:* (32) *is exact for quadratic paths, has a fourth-order error in general, and is provably rate-optimal under* (33)*.*

Representation (32) is how we implement (30): it requires *no* numerical differentiation step size, no explicit computation of curvatures, normals, or corner angles, and it automatically includes *every* term of (18) — both margin diagonals, both diagonal corner terms, and the cross corner term — because the functional itself is evaluated along the joint direction. It is a half-sample generalized jackknife: the correction subtracts the "curvature" of Θ between the two independent half-sample fits.

Remark 4.4 (The jackknife identity is not a curvature correction for non-smooth first stages)**.** *Lemma 4.2 is a statement about a smooth path. If $\hat{\gamma}^A, \hat{\gamma}^B$ are piecewise-constant (boosted trees, random forests), then $\phi(t) = \Theta(\bar{\gamma} + t\Delta)$ is a step function of t: it has no derivatives at all, $\phi^{(4)}$ does not exist, and Assumption 4.1 fails at every γ in the relevant neighborhood. Formula* (32) *remains a well-defined* estimator *— three evaluations of Θ — but it is no longer an approximation to $\frac{1}{8}D^2\Theta[\Delta, \Delta]$, and neither Theorem 4.6 nor Theorem 5.8 covers it. In Table 2 this mechanism is confounded with a second one: those cells also implement the superseded doubly debiased construction, whose own second-order defect (Remark 5.6) predicts over-correction on smooth designs too. The two can be separated only by re-running the corrected estimator of Definition 5.7 with smooth and tree first stages; see Section 6.5, item (L7).*

4.2 Isolating the corner: two families, two exact identities

Perturbing one contrast at a time isolates the corner. Because the decomposition can be carried out either on the closed-form $D^2\Theta$ or on finite differences, and because these do *not* agree exactly, we define both families and keep them separate. Write $C[u,v] := \int_{\mathcal{C}} uvw/(\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|)\,\mathrm{d}\mathcal{H}^{d-2}$ for the corner cross form, and recall from (18) the polarized identity $D^2\Theta[(u,v),(u,v)] = Q_g[u,u] + Q_h[v,v] + 2C[u,v]$.

Ideal family. With Q_g, Q_h, C evaluated at $\bar\gamma$, set

$$\hat\Theta_{SS}^{\mathrm{mar}} := \Theta(\bar\gamma) - \tfrac{1}{8}\big\{Q_g[\Delta_g,\Delta_g] + Q_h[\Delta_h,\Delta_h]\big\}, \qquad \text{corner} := \tfrac{1}{4}\,C[\Delta_g,\Delta_h]. \tag{34}$$

Then, *exactly*, $\hat\Theta_{SS} = \hat\Theta_{SS}^{\mathrm{mar}} - \text{corner}$, because $\frac{1}{8}D^2\Theta[\Delta,\Delta] = \frac{1}{8}(Q_g+Q_h) + \frac{1}{4}C$.

Jackknife family. With the second differences

$$\begin{aligned} q_{\mathrm{full}} &:= 4\big[\Theta(\hat\gamma^A) + \Theta(\hat\gamma^B) - 2\Theta(\bar\gamma)\big], \\ q_g &:= 4\big[\Theta(\hat g^A,\bar h) + \Theta(\hat g^B,\bar h) - 2\Theta(\bar\gamma)\big], \qquad q_h := 4\big[\Theta(\bar g,\hat h^A) + \Theta(\bar g,\hat h^B) - 2\Theta(\bar\gamma)\big], \end{aligned} \tag{35}$$

$$\hat\Theta_{SS}^{\mathrm{mar,jack}} := \Theta(\bar\gamma) - \tfrac{1}{8}(q_g+q_h), \qquad \widehat{\text{corner}} := \tfrac{1}{8}\,(q_{\mathrm{full}} - q_g - q_h),$$

one has, again *exactly*,

$$\hat\Theta_{SS}^{\mathrm{mar,jack}} - \widehat{\text{corner}} = \Theta(\bar\gamma) - \tfrac{1}{8}q_{\mathrm{full}} = \hat\Theta_{SS}^{\mathrm{jack}}. \tag{36}$$

Remark 4.5 (A correction to v3). *v3 asserted $\hat\Theta_{SS} = \hat\Theta_{SS}^{mar} - \widehat{corner}$, mixing the ideal left-hand side with the finite-difference right-hand side. That is false unless $t \mapsto \Theta(\bar\gamma + t\Delta)$ is exactly quadratic: by (36) the correct right-hand side is $\hat\Theta_{SS}^{\mathrm{jack}}$, and Lemma 4.2 puts $\hat\Theta_{SS}^{\mathrm{jack}} - \hat\Theta_{SS} = -\frac{1}{384}\phi^{(4)}(\xi)$. The two identities (34) and (36) are each exact within their own family; only the* cross-family *statement carries a fourth-order error. All statements below are labelled accordingly.*

Why q_g absorbs a corner term: perturbing g alone corresponds to the direction $(\Delta_g, 0)$, and $D^2\Theta[(\Delta_g,0),(\Delta_g,0)] = Q_g[\Delta_g,\Delta_g]$ *including* the $\cos\omega$ corner diagonal that (19) attaches to Q_g. Hence q_g, q_h absorb both corner diagonals and, by bilinearity,

$$q_{\mathrm{full}} - q_g - q_h = 2\,C[\Delta_g,\Delta_h] + O_p\big(\text{4th-order}\big), \tag{37}$$

the corner *cross* form alone. So $\widehat{\text{corner}}$ measures the cross term's stochastic diagonal, *not* the whole of (27). Identity (37) is verified numerically to full precision on thirteen closed-form designs in Appendix C; it matters because it determines exactly which part of the corner survives margins-only debiasing (Proposition 4.10).

Theorem 4.6 (SS debiasing: a conditional weak-smoothness result)**.** *Let Assumptions 1.4–1.8 and 4.1 hold, let the four arm regressions use sieve dimension $K_n \asymp n^a$ (up to logs), and suppose $s > (d+1)/2$. Define the* total *remainder of the corrected estimator,*

$$R_3^{SS} := \underbrace{\tfrac{1}{2}\{D^2\Theta(\gamma_\xi) - D^2\Theta(\gamma_0)\}[\bar{\Delta}, \bar{\Delta}]}_{\text{Taylor mean-value remainder}} - \underbrace{\tfrac{1}{8}\{D^2\Theta(\bar{\gamma}) - D^2\Theta(\gamma_0)\}[\Delta, \Delta]}_{\text{evaluation-point remainder}}, \tag{38}$$

both terms being bounded by the modulus (6). Then:

*(i) (*Rate, ideal estimator.*) At $a = d/(2s+1)$, if $R_3^{SS} = O_p(r_n^*)$ then $\hat{\Theta}_{SS} - \Theta_0 = O_p(r_n^*)$, $r_n^* = n^{-s/(2s+1)}$, which by Theorem 2.2(ii) is minimax-optimal.*

*(ii) (*Centered CLT, ideal estimator.*) If in addition $R_3^{SS} = o_p(\mathrm{se}_n)$ and*

$$\frac{d}{2s+1} \;<\; a \;<\; \min\Big\{\frac{d}{d+2},\; \frac{d}{3d-2}\Big\}, \tag{39}$$

a window nonempty iff *$s > (d+1)/2$ (known density, or $d = 2$, where the two edges coincide at $d/(d+2)$), respectively iff $s > 3(d-1)/2$ (empirical measure with $d \geq 3$), then $\sqrt{n}(\hat{\Theta}_{SS} - \Theta_0)/\hat{\sigma}_n \xrightarrow{d} \mathcal{N}(0,1)$ with $\hat{\sigma}_n$ from (15) under Proposition 2.4.*

*(iii) (*Centered CLT, jackknife estimator.*) For the tuning-free $\hat{\Theta}_{SS}^{\text{jack}}$ of (32), the fourth-order error of Lemma 4.2* must *also be $o(\mathrm{se}_n)$, which under Assumption 4.1(d) requires its own sieve window:*

$$\frac{d}{2s+1} \;<\; a \;<\; \min\Big\{\frac{3d}{4d+7},\; \frac{d}{3d-2}\Big\}, \tag{40}$$

nonempty iff $s > 2(d+1)/3$ (and $s > 3(d-1)/2$ in the empirical-measure case with $d \geq 3$). Since $3d/(4d+7) < d/(d+2)$ for every d, this is strictly inside (39).

Remark 4.7 (What v3 got wrong here)**.** *Two things.* (a) *v3 stated part (iii) as "the same conclusions hold under the additional* smoothness *condition (33)". But (33) was derived by comparing the fourth-order error with r_n^* at the rate-optimal K_n; a centred CLT compares it with se_n at an* undersmoothed *K_n, which is a different and a-dependent requirement. Without (40) one may choose $a \in (3d/(4d+7), d/(d+2))$ satisfying every stated condition while the jackknife error dominates the standard error.* (b) *v3's R_3 was the Taylor remainder only. The estimator subtracts $D^2\Theta(\bar{\gamma})[\Delta, \Delta]$ while the cancellation argument uses $D^2\Theta(\gamma_0)$; the evaluation-point difference in (38) is a separate third-order term of the same order, not covered by the Taylor remainder. Both are now bounded by the single modulus (6).*

Remark 4.8 (R_3^{SS} is an assumption, not a consequence)**.** *Assumption 1.8(a)–(b) is where the weak-smoothness claim actually rests, and we have not verified it from primitive conditions. What can be said is how restrictive it is under each candidate modulus ϖ. Substituting $\|\bar{\Delta}\|_\infty \asymp \sqrt{K_n/n}$ and $\|\bar{\Delta}\|_{C^1} \asymp K_n^{1/2+1/d} n^{-1/2}$ at $a = d/(2s+1)$ into $R_3^{SS} \lesssim \varpi(\|\bar{\Delta}\|_{C^1}) \|\bar{\Delta}\|_\infty \|\bar{\Delta}\|_{C^1}$:*

$$\varpi(\epsilon) \asymp \epsilon \text{ in } \|\cdot\|_{C^1} \quad \text{(2 gradients)} \implies s > \tfrac{3d+1}{4},$$
$$\varpi(\epsilon) \asymp \epsilon \text{ in } \|\cdot\|_\infty \quad \text{(1 gradient)} \implies s > \tfrac{3d-1}{4},$$

against the crude $R_3 \lesssim \|\bar{\Delta}\|_{C^1}^3$ of v3, which gives $s > 3(d+1)/4$. Only the last is weaker than $(d+1)/2$ at $d = 2$ (1.25 against 1.5). Since $D^2\Theta$ in (18)–(19) carries exactly one gradient across its two directions, the one-gradient modulus is the plausible one, and the same conjecture would close the jackknife gap of Remark 4.3. Establishing it requires a Lipschitz bound on $\gamma \mapsto D^2\Theta(\gamma)$ — which, unlike a third shape derivative, does not *require extra smoothness of γ_0 and is therefore compatible with the Hölder parameter space for every $s > 1$. That is the principal technical task left open by this draft. Until it is done, Theorem 4.6 is conditional: $s > (d+1)/2$ is the range on which* every other *term in the error budget is controlled, and Assumption 1.8(b) is imposed on that range.*

Proof sketch. Expand $\Theta(\bar{\gamma})$ around γ_0 to second order and substitute (31). The linear term $D\Theta[\bar{\Delta}]$ is the average of two independent half-sample linear terms and delivers the CLT with the same σ_n. The quadratic term decomposes as

$$\tfrac{1}{2}D^2\Theta[\bar{\Delta},\bar{\Delta}] - \tfrac{1}{8}D^2\Theta[\Delta,\Delta] = \underbrace{\tfrac{1}{2}D^2\Theta[b,b]}_{\text{(i)}} + \underbrace{D^2\Theta[b,\bar{u}]}_{\text{(ii)}} + \underbrace{\tfrac{1}{2}D^2\Theta[u^A,u^B]}_{\text{(iii)}},$$

the stochastic diagonals having cancelled exactly. By Table 1: (i) is $O(K_n^{-(2s-1)/d}) = o(r_n^*)$ under $s > 1$; (ii) is $O_p(K_n^{(1-s)/d}\sqrt{K_n/n})$; and (iii) is a mean-zero degenerate U-statistic of order $n^{-1}K_n^{(d+3)/(2d)}$ (Appendix B). For (i), at $a = d/(2s+1)$ both (ii) and (iii) are $O_p(r_n^*)$ iff $s \geq (d+1)/2$. For (ii), relative to $\sqrt{K_n^{1/d}/n}$ the term (ii) carries exponent $a(d-2s+1)/(2d)$, negative for *every* $a > 0$ iff $s > (d+1)/2$ — so it is a pure smoothness condition — while (iii) is $o(\sqrt{K_n^{1/d}/n})$ iff $a < d/(d+2)$, and the sieve bias is $o(\sqrt{K_n^{1/d}/n})$ iff $a > d/(2s+1)$; the two bracket (39), nonempty iff $2s+1 > d+2$. Thus *the binding constraint $s > (d+1)/2$ comes from the margin terms (ii) and (iii)*; the corner analogues of both are smaller and impose no binding requirement beyond transversality. The remaining pieces are R_3^{SS} of (38), assumed negligible by Assumption 1.8(b) — see Remark 4.8 — the equicontinuity term of Assumption 1.8(c), and, for the jackknife implementation, the fourth-order error of

Lemma 4.2, whose own window is (40).

Ratio consistency of $\hat\sigma_n$ follows from Proposition 2.4 *not* because the correction is small — inside (39) it need not be, and v3's assertion that it is $o_p(\sigma_n/\sqrt{n})$ was wrong — but because the above displays an *asymptotically linear* representation $\hat\Theta_{SS}-\Theta_0 = \frac{1}{n}\sum_i\{v_S^*\epsilon_{S,i}+v_Y^*\epsilon_{Y,i}\}+o_p(\mathrm{se}_n)$ with the *same* influence function as the plug-in. The whole point of the estimator is that the correction cancels the plug-in's quadratic diagonal; both may exceed se_n individually while their sum does not. ☐

Remark 4.9 (What "$s>(d+1)/2$" is, and is not). *v1 asserted that at the rate-optimal K_n the condition $s>(d+1)/2$ is "exactly" $K_n^{1/2-s/d}\to 0$. That is false: $K_n^{1/2-s/d} = n^{(d-2s)/(2(2s+1))}\to 0$ iff $s>d/2$. The correct equivalent statement in terms of the estimated geometry is*

$$\frac{K_n^{1/2+1/d}}{\sqrt{n}}\longrightarrow 0 \qquad\Longleftrightarrow\qquad s>\frac{d+1}{2}\quad \textit{at } K_n\asymp n^{d/(2s+1)}, \tag{41}$$

i.e. uniform consistency of the gradient *of the estimated surface (each derivative costing $K_n^{1/d}$ on top of the $\sqrt{K_n/n}$ pointwise stochastic size). It is not a coincidence that the same threshold appears in the proof of Theorem 4.6: both terms (ii) and (iii) there carry exactly one gradient factor.*

Proposition 4.10 (Margins-only debiasing is insufficient). *Under the conditions of Theorem 4.6,* within each family separately,

$$\hat\Theta_{SS}^{mar}-\hat\Theta_{SS} = corner\quad (ideal,\ (34)),\qquad \hat\Theta_{SS}^{mar,jack}-\hat\Theta_{SS}^{\mathrm{jack}} = \widehat{corner}\quad (jackknife,\ (36)),$$

both exact; the cross-family statement carries the fourth-order error of Lemma 4.2. By (37) the error of the margins-only estimator retains the corner cross *term and nothing else of the corner, with expectation $\mathcal{B}^{\times}_{\mathcal{C},n}$ of (28) splitting as*

$$\mathcal{B}^{\times}_{\mathcal{C},n} = \underbrace{\mathcal{B}^{\times,\mathrm{stoch}}_{\mathcal{C},n}}_{O(K_n/n)} + \underbrace{\mathcal{B}^{\times,\mathrm{bias}}_{\mathcal{C},n}}_{O(K_n^{-2s/d})=O(r_n^{*2})}.$$

The two $\cos\omega$ *corner diagonals of (19) are components of Q_g and Q_h and* are *removed by the margins-only correction. The split-sample statistic $\widehat{\mathrm{corner}}$ is unbiased for $\mathcal{B}^{\times,\mathrm{stoch}}_{\mathcal{C},n}$ only: because the common approximation bias cancels from $\Delta=\hat\gamma^A-\hat\gamma^B$, $\mathbb{E}[\widehat{\mathrm{corner}}]$ contains no bias-product term. The bias part is $o(r_n^*)$ and is removed by undersmoothing, not by debiasing. Consequently,* provided *Assumption 1.6(c2) holds — which neither (c1) nor pointwise nonvanishing of ϱ_{gh} implies, the three being logically independent restrictions — $\hat\Theta_{SS}^{mar}$ carries* *a residual bias $\asymp K_n/n$, which is* not $o(r_n^*)$ *when $s \leq d-1$; since the SS guarantee of Theorem [4.6](#) extends down to $s > (d+1)/2$, margins-only debiasing forfeits the weak-smoothness guarantee on the window (26), nonempty whenever $d \geq 4$. If the two contrast errors are conditionally uncorrelated at every corner, margins-only and corner-aware debiasing agree to this order. For $d \in \{2, 3\}$ the window is empty and the corner cross term is a finite-sample constants effect — of order at most $K_n^{-1/d}$ times the margin diagonals' bound, and by Remark 3.4 plausibly of the* same *order as the margin diagonals actually are on a generic geometry.*

Proof. The first display is the definition of $\widehat{\text{corner}}$ in (35). For the second, apply the decomposition in the proof of Theorem 4.6 to

$$D^2\Theta[\bar{\Delta}, \bar{\Delta}] - \tfrac{1}{4}D^2\Theta[(\Delta_g, 0), (\Delta_g, 0)] - \tfrac{1}{4}D^2\Theta[(0, \Delta_h), (0, \Delta_h)] :$$

by bilinearity the Q_g and Q_h stochastic diagonals cancel in full — including their corner components — leaving $C[\bar{\Delta}_g, \bar{\Delta}_h]$, whose expectation is $\mathcal{B}_{\mathcal{C}}^{\times}$. Splitting $\bar{\Delta} = b + \bar{u}$ and using $\mathbb{E}[\bar{u}] = 0$ separates it into the stated stochastic and bias parts, of orders $|\varrho_{gh}|\, K_n/n$ and $K_n^{-2s/d}$ by (24). That $\mathbb{E}[\widehat{\text{corner}}]$ omits the bias part is immediate from $\Delta = u^A - u^B$ in (31). $\square$

Remark 4.11 (Relation to v1)**.** *v1's Proposition on margins-only debiasing stated $\mathbb{E}[\widehat{corner}] \asymp 2C[b_g, b_h] + \rho_{SY} K_n/n$. The identification of the leftover with the ρ_{SY}-driven* cross *term is correct — an earlier revision of the present draft wrongly reassigned the $\cos\omega$ diagonals to it, and (37) settles the matter. What does not belong is the bias-product term in $\mathbb{E}[\widehat{corner}]$, whose presence contradicted v1's own (correct) description of Table 4. Proposition 4.10 keeps the two objects separate: the* residual bias *of $\hat{\Theta}_{SS}^{mar}$ contains both parts; the* SS estimate of it *targets only the stochastic part. The empirical "remainder" column of Table 4 is precisely $\mathcal{B}_{\mathcal{C}}^{\times,\text{bias}}$, and its near-constancy in n at fixed K is the corresponding prediction.*

The pair (Theorem 4.6, Proposition 4.10) is the central methodological message of the two-threshold extension: *quadratic debiasing must be corner-aware.* Because the jackknife form (32) perturbs both contrasts jointly, it is corner-aware by construction — the theory identifies what would go wrong with the seemingly natural surface-by-surface application of the single-threshold correction.

Remark 4.12 (Leave-one-out analogue)**.** *The LOO estimator replaces $\frac{1}{8}D^2\Theta[\Delta, \Delta]$ by*

$$\frac{1}{2n^2} \sum_{i=1}^{n} D^2\Theta(\hat{\gamma})\big[(s_i\hat{\epsilon}_{S,i}^{(-i)},\, s_i\hat{\epsilon}_{Y,i}^{(-i)}),\ (s_i\hat{\epsilon}_{S,i}^{(-i)},\, s_i\hat{\epsilon}_{Y,i}^{(-i)})\big],$$

whose corner cross component carries the product $\hat{\epsilon}_{S,i}^{(-i)}\hat{\epsilon}_{Y,i}^{(-i)}$ *— an explicit estimate of the* ρ_{SY}*-weighted part of* (27) *— while its corner diagonal components carry* $(\hat{\epsilon}_{S,i}^{(-i)})^2$ *and* $(\hat{\epsilon}_{Y,i}^{(-i)})^2$*, which is why the LOO correction, like the SS correction, captures the* $\cos\omega$ *terms as well. We expect it to satisfy the analogue of Theorem 4.6 under the corresponding LOO conditions of Chen and Gao (2026), but we do not claim it: the LOO construction carries its own* R_{diag} *remainder, distinct from* (38)*, which we have not analysed. We focus on SS in the simulations because* (32) *is tuning-free.*

5 Sieve DML and the Doubly Debiased Estimator

5.1 Cross-fitted two-band sieve-Riesz correction

The debiasing of Section 4 targets the *second-order* remainder but retains sieve LS first stages. The complementary device — the sieve-Riesz DML of the extended single-threshold draft — targets the *first-order* term on the sieve space and thereby accommodates generic machine-learning first stages. Let $(I_\kappa)_{\kappa=1}^K$ partition the sample, $\hat{\mu}_{-\kappa}$ be *any* first-stage learner of the four arm means fit off fold κ, and $\hat{v}^*_{K_n,o,-\kappa}$ the feasible two-band sieve-Riesz representer (12)–(13) computed *off fold* κ — both its Gram matrix and its bands must use only $I_{-\kappa}$. The cross-fitted sieve-debiased estimator is

$$\tilde{\Theta} := \frac{1}{K}\sum_{\kappa=1}^{K}\left[\Theta(\hat{\gamma}_{-\kappa}) + \frac{K}{n}\sum_{i\in I_\kappa}\sum_{o\in\{S,Y\}} \hat{v}^*_{K_n,o,-\kappa}(X_i,W_i)\big(O_i - \hat{\mu}_{o,-\kappa}(X_i,W_i)\big)\right], \tag{42}$$

with $O_i \in \{S_i, Y_i\}$. The correction annihilates the first-order estimation error of $\hat{\mu}_{-\kappa}$ *on the sieve space*: it is the local influence function of the irregular functional, well defined even though no global influence function exists.

Assumption 5.1 (Two-band local robustness)**.** *For each fold* κ*: (a)* $\|\hat{\mu}_{-\kappa}-\mu_0\|_\infty = o_p(1)$ *and* $\|\hat{\gamma}_{-\kappa}-\gamma_0\|_\infty = o_p(\varepsilon_n)$*; (b) the band sequence satisfies* $\varepsilon_n \downarrow 0$*,* $n\varepsilon_n/(K_n\log K_n)\to\infty$*, and* $\varepsilon_n^{-1}\|\hat{\gamma}_{-\kappa}-\gamma_0\|_\infty = o_p(1)$*; (c) (*boundary stability, normal-graph form*) for both* $\varphi\in\{g,h\}$ *with* $\mathcal{M}_\varphi := \{\varphi_0 = 0\}$ *nonempty, the estimated level set admits, w.p.* $\to 1$*, the normal-graph representation*

$$\{\hat{\varphi}_{-\kappa}=0\} = \big\{x + r_n(x)\,n_\varphi(x) \;:\; x\in\mathcal{M}_\varphi\big\}, \qquad \|r_n\|_{C^1(\mathcal{M}_\varphi)} = o_p(1), \tag{43}$$

with $\mathcal{M}_\varphi$ *of positive reach and the graph map's Jacobian bounded above and below uniformly; if* $\mathcal{M}_\varphi=\emptyset$ *we require instead* $\mathbb{P}(\{\hat{\varphi}_{-\kappa}=0\}\neq\emptyset)\to 0$*. Condition* (43) *implies* $d_H\to 0$ *and* $\mathrm{Leb}(\{\hat{\varphi}\geq 0\}\triangle\{\varphi_0\geq 0\})\to 0$*, but is* strictly stronger *than both together (Remark 5.2);*

(d) the product rate $\left\|\hat{v}^*_{K_n,-\kappa} - v^*_{K_n}\right\| \cdot \|\hat{\mu}_{-\kappa} - \mu_0\| = o_p(\sqrt{K_n^{1/d}/n})$*; (d′) (*representer consistency*)* $\left\|\hat{v}^*_{K_n,-\kappa} - v^*_{K_n}\right\|_{sd} = o_p(\sigma_n)$*; (e) the sieve approximation bias of the representer is* $o(\sqrt{K_n^{1/d}/n})$*, and the out-of-span boundary remainder* R_5 *of* (44) *is* $o_p(\sqrt{K_n^{1/d}/n})$*; (f) (*leading nonlinear remainder*) writing* $\Delta := \hat{\gamma}_{-\kappa} - \gamma_0$*,*

$$\|\Delta\|_\infty \|\Delta\|_{C^1} + \|\Delta\|_\infty^2 = o_p\big(\sqrt{K_n^{1/d}/n}\big),$$

which bounds $|D^2\Theta[\Delta, \Delta]|$ *by* (18)–(19)*, together with a genuine third-order remainder condition* $R_3 = o_p(\sqrt{K_n^{1/d}/n})$*; (g) the Lyapunov condition Assumption efass:rem(d) holds, and* $\liminf \sigma_n^2/K_n^{1/d} > 0$*, the latter supplied by Assumption efass:nondegen(b).*

Remark 5.2 (Two corrections to v2's assumption list)**.** *Condition (c) as stated in v2 was* $\mathcal{H}^{d-1}(\{\hat{g} = 0\}\triangle\{g_0 = 0\}) = o_p(1)$*. That is not merely strong, it is unsatisfiable: two* distinct *hypersurfaces have essentially disjoint* $\mathcal{H}^{d-1}$*-supports, so the symmetric difference has measure* $\approx 2\mathcal{H}^{d-1}(\{g_0 = 0\})$ *no matter how close they are — two parallel segments* 10^{-10} *apart already fail it. v2 also stated (c) for g only. v3 replaced it by Hausdorff distance* plus *region symmetric-difference measure and called the pair "equivalent" to normal-graph convergence. That is also wrong, in the opposite direction: a surface can be arbitrarily Hausdorff-close to* $\mathcal{M}_\varphi$ *while oscillating at a fine scale, so that its surface measure diverges and its normals do not converge, all with region symmetric-difference measure tending to zero. Since the band construction* (13) *converges to a* Hausdorff *integral — carrying* $\|\nabla g_0\|^{-1}$ *and hence the normal direction — neither metric controls what is needed. Hence (c) now* requires *the normal-graph representation* (43) *with* C^1 *control of the displacement, which is what makes the coarea change of variables valid with a Jacobian tending to one; the empty-margin case needs a separate convention because Hausdorff distance to* $\emptyset$ *is not defined. Condition (f) in v2 read* $\|\Delta\|_{C^1}^2 \|\Delta\|_\infty$ *and called it the "second-order remainder"; under* $\Delta = t\eta$ *that quantity is* $O(t^3)$*, whereas the term it is supposed to control,* $\frac{1}{2}D^2\Theta[\Delta, \Delta]$*, is generically* $O(t^2)$*. A cubic bound cannot control a quadratic term, so Theorem 5.3 did not follow from v2's list. The corrected (f) has the right homogeneity, and for a sieve LS first stage it reduces to* $K_n^{1+1/d}/n$*, consistent with Table 1. Condition (d′) is new: the product rate in (d) can hold with an oracle learner (*$\|\hat{\mu} - \mu_0\| = 0$*) for an arbitrarily wrong* $\hat{v}^*$*, which would still ruin* $\hat{\sigma}_n$*.*

Theorem 5.3 (Two-band sieve DML)**.** *Under Assumptions efass:model– efass:rem and efass:dml,*

$$\frac{\sqrt{n}\,(\tilde{\Theta} - \Theta_0)}{\sigma_n} \xrightarrow{d} \mathcal{N}(0, 1), \qquad \sigma_n^2 \asymp K_n^{1/d},$$

and the two-band variance estimator (15)*, computed with cross-fitted residuals, is ratio-*

consistent. As in the single-threshold case, the irregularity weakens *the product-rate requirement relative to regular DML: the tolerance is* $\sqrt{K_n^{1/d}/n} \gg n^{-1/2}$.

Assumption 5.1 is longer than the corresponding list in v1, which cited only sup-norm consistency, a product rate, and sieve approximation bias. Those three do not control a boundary functional: (b) and (c) are what make the ε-band a consistent surrogate for a Hausdorff integral at an *estimated* surface, (d′) is what makes the variance estimate consistent, (f) is the nonlinear (moving-boundary) remainder that has no analogue in regular DML, and (g) is what a CLT for a triangular array with $\sigma_n^2 \to \infty$ requires. We state them because the two conditions that bind in our experiments are (e) and (f), not the product rate. As in Theorem 2.2(iii), the conclusion is a *centered* limit only because (e) forces the representer's approximation bias below $\sqrt{K_n^{1/d}/n}$, which is again an undersmoothing requirement.

Remark 5.4 (Least-squares orthogonality; why raw AIPW corrections miscalibrate)**.** *Suppose the first stage* is *the sieve LS estimator on the same basis used for the representer, and consider the* non-cross-fitted *estimator: a full-sample fit with in-sample residuals. Then the normal equations* $\sum_i \psi(X_i, W_i)\hat{\epsilon}_i = 0$ *hold exactly, and since* $\hat{v}^*$ *lies in the span of* ψ *the correction vanishes identically: the sieve plug-in is automatically sieve-debiased at first order. This explains a phenomenon documented in our earlier experiments: adding a* raw *ε-band AIPW correction* $(2\varepsilon)^{-1}\mathbb{1}\{|\hat{\tau}| < \varepsilon\}\xi_i$ *on top of a full-sample sieve plug-in over-corrects — the projected (sieve-Riesz) correction it approximates is exactly zero, and the discrepancy is pure noise with a* $1/\varepsilon$ *variance inflation.*

This exactness does not survive cross-fitting*, contrary to what v1 asserted. In* (42) *the residuals* $O_i - \hat{\mu}_{-\kappa}(X_i)$ *for* $i \in I_\kappa$ *are* out-of-fold*: the normal equations hold on* $I_{-\kappa}$*, not on* I_κ*, so the correction is a nondegenerate mean-zero term of the same order* $\sqrt{K_n^{1/d}/n}$ *as the leading variance. With a sieve LS first stage the cross-fitted correction is therefore not zero but merely* redundant*: it substitutes one* $O_p(\sqrt{K_n^{1/d}/n})$ *term for another without changing the limit distribution. The projected correction* (42) *is* useful *precisely when the first stage is not the LS fit on the representer's basis — e.g. gradient boosting or random forests — which is its intended use case.*

Remark 5.5 (When the sieve cannot see the learner's error)**.** *Assumption 5.1(e) deserves emphasis because it is the one that binds in practice. Write* $\Delta := \hat{\mu}_{-\kappa} - \mu_0$ *and decompose the plug-in's first-order error as*

$$D_\mu\Theta(\mu_0)[\Delta] = \underbrace{\left\langle v^*_{K_n}, \Delta \right\rangle}_{\textit{in-span}} + \underbrace{D_\mu\Theta(\mu_0)\big[(I - \Pi_{K_n})\Delta\big]}_{=:R_5,\ \textit{out-of-span}}. \tag{44}$$

The correction in (42) *replaces the first term by the score* $\mathbb{E}_n[v^*\epsilon]$*;* R_5 *is untouched. If* $\hat{v}^*_{-\kappa}$ *and* $\hat{\mu}_{-\kappa}$ *are measurable with respect to* $I_{-\kappa}$ *and the score is conditionally mean zero, then* $\mathrm{Cov}(\mathit{score}, R_5) = 0$ *exactly, so*

$$\mathrm{Var}(\tilde{\Theta}) = \frac{\sigma_n^2}{n} + \mathrm{Var}(R_5) + \mathrm{Var}(h.o.t.) + \textit{cross terms in the h.o.t.},$$

while $\hat{\sigma}_n^2$ *estimates only* σ_n^2*.* This is a statement about which terms the standard error omits; it is not a statement about the sign of the coverage error*, and v1's conclusion that the interval "can only under-cover, never over-cover" does not follow, for three reasons. First, the plug-in error* also *contains* R_5*; the net change in dispersion from adding the correction is* $\sigma_n^2/n - \mathrm{Var}(\langle v^*_{K_n}, \Delta\rangle)$*, of either sign, so the debiased estimator is not "the plug-in plus an independent perturbation" except in the degenerate case* $\Pi_{K_n}\Delta \equiv 0$*. Second, coverage depends on bias as well as dispersion, and* $\hat{\sigma}_n$ *is itself a noisy band-based estimate whose own bias — upward at small* ε_n*, downward when the band is widened — is not sign-restricted; SE/SD ratios above one occur in several cells of Table 2. Third, the exact-orthogonality step requires the representer to be built off fold* κ*, which our current implementation does not do (Section 6.5, item (L6)).*

Directly measured, the variance-addition channel is small on our design: the saved variance experiment reports $\{\mathrm{Var}(\mathrm{DML}) - \mathrm{Var}(\text{plug-in})\}/\widehat{\mathrm{SE}}^2$ *equal to* -0.04 *(GBR),* -0.00 *(RF) and* $+0.04$ *(smooth RBF ridge) at* $n = 4000$ *— not the* $\approx +1$ *that the "independent added variance" reading predicts.* The machine-learning undercoverage on this design is therefore predominantly residual bias, not unmodelled variance. *We correct v1, which asserted the opposite.*

What must be small in Assumption 5.1(e) is not $\|(I - \Pi_{K_n})\Delta\|_{L^2}$ *but the* boundary *functional evaluated at it: only the sieve's ability to track the learner's error* on the margins *matters, and global* L^2 *richness is not the right diagnostic. This is exactly where regression trees fail. A boosted ensemble's error is piecewise constant with jumps at adaptively chosen split hyperplanes, and boosting places its splits where the loss gradient is largest — for a CATE surface whose zero level set defines the estimand, that is near the margins themselves. So* $(I - \Pi_{K_n})\Delta$ *is largest exactly on the set the derivative integrates over. Section 6 measures a global-*L^2 *proxy for this: the* R^2 *of the boosted first-stage error on the tensor B-spline representer span is* 0.21–0.34 *across the four arm regressions (mean* 0.26*), and the projected correction removes about* 31% *of the plug-in's boundary bias. Tree ensembles additionally violate the smoothness conditions in a way that is not merely quantitative:* $\nabla(\hat{\mu} - \mu_0)$ *is a sum of Dirac measures on the split hyperplanes, the* ε*-band* $\{|\hat{\tau}| < \varepsilon\}$ *is a union of whole leaf cells (of measure* $O(1)$ *or exactly zero, never* $O(\varepsilon)$*), and the estimated region is an axis-aligned* *staircase whose surface measure over-counts an oblique boundary by up to $\sqrt{d}$. The practical conclusion is stated in Section 6.4: match the representer basis to the learner's own feature map, and prefer smooth learners.*

The diagnosis is not special to two thresholds. In the single-threshold setting, the extended draft's own Table 7 reports, for the value functional on identical samples and folds, a gradient-boosting DML estimator that is essentially *unbiased* ($|\text{bias}| \leq 0.003$ at every n) yet covers only 0.90, with $\text{SE/SD} = 0.78, 0.84, 0.88$ — while a smooth random-feature first stage on the same design has $\text{SE/SD} = 0.84, 0.94, 0.99$ and coverage rising to 0.93. The decomposition here explains that contrast, though — given the variance measurement above — the mechanism in the two-threshold design is the omitted-bias channel rather than the omitted-variance channel.

5.2 The doubly debiased estimator

The two devices remove different terms — the first-order projection error (DML) and the second-order diagonal (SS) — and one would like to compose them. Doing so naively does not work, and we begin by saying why, because the construction proposed in v1–v2 of this draft is invalid.

Split the sample into halves A, B; fit generic ML first stages on each half; let $\hat{\gamma}^A, \hat{\gamma}^B$ be the two contrast pairs (each predicted on the full covariate sample), $\bar{\gamma}$ their average, and $e_j := \hat{\gamma}^j - \gamma_0$. The v1–v2 estimator added ordinary fold-specific one-step corrections to the split-sample combination:

$$\hat{\Theta}_{DD}^{\text{v2}} := \underbrace{2\,\Theta(\bar{\gamma}) - \tfrac{1}{2}\big(\Theta(\hat{\gamma}^A) + \Theta(\hat{\gamma}^B)\big)}_{=:\ \Psi(\hat{\gamma}^A, \hat{\gamma}^B)} + \frac{1}{n}\sum_{i=1}^{n}\sum_{o\in\{S,Y\}} \hat{v}^*_{K_n,o,-\kappa(i)}(X_i, W_i)\,\hat{\epsilon}^{\text{oof}}_{o,i}. \tag{45}$$

Remark 5.6 ((45) reintroduces the diagonals it was meant to remove)**.** *Expanding Ψ around γ_0 and using $\bar{e} = \frac{1}{2}(e_A + e_B)$,*

$$\Psi(\hat{\gamma}^A, \hat{\gamma}^B) - \Theta_0 = D\Theta[\bar{e}] + \tfrac{1}{2}D^2\Theta[e_A, e_B] + O(\mathit{3rd}), \tag{46}$$

the own-half diagonals having cancelled — this is the whole point of SS. But the correction in (45) uses representers built at $\hat{\gamma}^A, \hat{\gamma}^B$, so its conditional expectation given the fits is

$$-\tfrac{1}{2}D\Theta(\hat{\gamma}^A)[e_A] - \tfrac{1}{2}D\Theta(\hat{\gamma}^B)[e_B] = -D\Theta(\gamma_0)[\bar{e}] - \tfrac{1}{2}\big(D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]\big) + O(\mathit{3rd}).$$

The linear terms cancel as intended, but the second line reintroduces the own-half diagonals, *at $-\frac{1}{2}$ each. The cleanest way to see it: take $e_A = e_B = b$ deterministic. Then $\bar\gamma = \hat\gamma^A = \hat\gamma^B$, $\Psi = \Theta(\gamma_0 + b)$ — SS does nothing, correctly, since there is no independent replicate — and (45) has error $\frac{1}{2}D^2\Theta[b,b] - D^2\Theta[b,b] = -\frac{1}{2}D^2\Theta[b,b]$, which is* quadratic*, not third order. With independent mean-zero half-sample errors the bias is $-\frac{1}{2}\mathbb{E}\{D^2\Theta[e_A,e_A] + D^2\Theta[e_B,e_B]\} \asymp K_n^{1+1/d}/n$ — worse than the plug-in's, and with the opposite sign. Theorem "DD" of v1–v2 is therefore false as stated. Appendix C, item 7, exhibits both facts numerically on the lens design.*

The error is that (45) debiases the wrong functional. Ψ is a function of *two* arguments, and its derivative in the first is

$$D_A\Psi = D\Theta(\bar\gamma) - \tfrac{1}{2}D\Theta(\hat\gamma^A), \qquad \text{not} \qquad \tfrac{1}{2}D\Theta(\hat\gamma^A). \tag{47}$$

The two agree to first order — both are $\approx \frac{1}{2}D\Theta(\gamma_0)$ — and differ exactly at the order that matters. Substituting (47) restores the intended cancellation: a short computation gives $D_A\Psi[e_A] = \frac{1}{2}D\Theta_0[e_A] + \frac{1}{2}D^2\Theta[e_A,e_B]$ and symmetrically, so with (46)

$$\Psi - \Theta_0 - D_A\Psi[e_A] - D_B\Psi[e_B] = -\tfrac{1}{2}D^2\Theta[e_A,e_B] + O(\text{3rd}), \tag{48}$$

a mean-zero cross-half term of the same order as SS's own leftover.

Implementing (47) requires care: $\bar\gamma$ depends on *both* halves, so no observation used to fit either is available for a conditionally-mean-zero score. We therefore use three folds.

Definition 5.7 (Corrected doubly debiased estimator)**.** *Partition the sample into three folds I_1, I_2, I_3 of equal size. For each cyclic assignment π of (A,B,C) to (I_1,I_2,I_3): fit $\hat\gamma^A$ on A and $\hat\gamma^B$ on B; form $\bar\gamma$; build the two feasible sieve-Riesz representers $\hat v^{*A}, \hat v^{*B}$ of the composite derivatives $D\Theta(\bar\gamma) - \frac{1}{2}D\Theta(\hat\gamma^A)$ and $D\Theta(\bar\gamma) - \frac{1}{2}D\Theta(\hat\gamma^B)$ via (12)–(13), using only $A \cup B$ data for the bands and the Gram matrix; and set*

$$\hat\Theta_{DD}^{(\pi)} := \Psi(\hat\gamma^A, \hat\gamma^B) + \frac{3}{n}\sum_{i\in C}\sum_{o\in\{S,Y\}}\sum_{j\in\{A,B\}} \hat v_o^{*j}(X_i,W_i)\big(O_i - \hat\mu_o^j(X_i,W_i)\big). \tag{49}$$

The corrected DD estimator is $\hat\Theta_{DD} := \frac{1}{3}\sum_\pi \hat\Theta_{DD}^{(\pi)}$ over the three cyclic assignments.

Because $\hat v^{*A}, \hat v^{*B}, \hat\mu^A, \hat\mu^B$ are all $A \cup B$-measurable while the score runs over C, the score is conditionally mean zero and its conditional expectation is $-\langle \hat v^{*A}, e_A\rangle - \langle \hat v^{*B}, e_B\rangle \approx -D_A\Psi[e_A] - D_B\Psi[e_B]$, which is exactly what (48) requires.

Two definitional points that Definition 5.7 must settle, and which v3 left open. *Fold conventions for Ψ.* In the empirical-measure case $\Theta(\cdot)$ is itself a sample average, so one must say over which observations. We evaluate every $\Theta(\cdot)$ inside Ψ over the *full* sample. The resulting discrepancy from the population Θ is the equicontinuity term of Assumption 1.8(c), already assumed $o_p(\mathrm{se}_n)$; it is common to all three evaluations in Ψ and therefore does not disturb the cancellation in (48), but it is *not* negligible by fiat and is the reason (c) is stated for every estimator in this paper rather than for the plug-in alone. In the known-density case the issue does not arise. *Representer conditions.* Assumption 5.1(d),(d′),(e) must be read for the *composite* representers: $\hat{v}^{*A}$ is required to satisfy them as an estimator of the Riesz representer of $D_A\Psi = D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^A)$, not of $\frac{1}{2}D\Theta(\gamma_0)$. Since both terms are band constructions at C^1-close level sets, the conditions transfer, but they are conditions on a *difference* of band vectors and we state them as such rather than inheriting them.

Three practical points. *(i)* A single assignment scores on one third of the sample, so its score variance is $3\sigma_n^2/n$; but $\frac{1}{3}\sum_\pi \frac{3}{n}\sum_{i\in C_\pi}(\cdot) = \frac{1}{n}\sum_{i=1}^n(\cdot)$ over the three cyclic assignments, so the rotation recovers the full-sample score and σ_n^2/n, as in ordinary cross-fitting. *(ii)* The fits use $n/3$ rather than $n/2$ observations, inflating the second-order terms by a constant; this is the price of the extra fold and does not affect rates. *(iii)* $\hat{v}^{*A}$ is built from a *difference* of two band vectors, at $\bar{\gamma}$ and at $\hat{\gamma}^A$. Since $D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^A) \approx \frac{1}{2}D\Theta(\gamma_0)$ the difference is not a near-cancellation, so the construction is numerically benign; but the two band vectors must be computed at genuinely different level sets, which is the step an implementation is most likely to short-cut.

Theorem 5.8 (Doubly debiased estimation, corrected)**.** *Suppose Assumption 5.1 holds for the ML first stages with (f) replaced by its third-order half, and in addition: (i) Assumption 4.1 holds along $\bar{\gamma} + t\Delta$, $|t| \le \frac{1}{2}$, w.p. $\to 1$; (ii) $\|\hat{\gamma}^j - \gamma_0\|_{C^1} = o_p(1)$ and $\hat{\gamma}^A$, $\hat{\gamma}^B$ are independent; (iii) the third-order margin/corner remainders and the fourth-order jackknife error of Lemma 4.2 are $o_p(\sqrt{K_n^{1/d}/n})$; (iv) $D^2\Theta[e_A, e_B] = o_p(\sqrt{K_n^{1/d}/n})$. Then for $\hat{\Theta}_{DD}$ of Definition 5.7,*

$$\frac{\sqrt{n}\,(\hat{\Theta}_{DD} - \Theta_0)}{\hat{\sigma}_n} \xrightarrow{d} \mathcal{N}(0,1),$$

with $\hat{\sigma}_n$ from (15) on cross-fitted residuals. The variance estimator must be formed from the actual *composite score: $\hat{\sigma}_n^2 = \frac{3}{n}\sum_{i\in C}\{\sum_o \sum_j \hat{v}_o^{*j}(X_i, W_i)\hat{\epsilon}_{o,i}^j\}^2$, averaged over the three assignments — not the ordinary two-band score of (15), unless the two are shown to be ratio-equivalent. (They should be, since $\hat{v}^{*A} + \hat{v}^{*B} \approx v_{K_n}^*$, but we do not prove it.)*

The gain over Theorem 5.3 is precise but narrower than v3 claimed: the quadratic correction replaces $D^2\Theta[\Delta,\Delta]$ — whose expectation contains the own-half stochastic diagonal, $\asymp K_n^{1+1/d}/n$ for a sieve — by $D^2\Theta[e_A, e_B]$, in which that diagonal is absent. Condition (iv) is itself a second-order condition*, so it is wrong to describe the requirement as dropping "from second to third order"; what drops is the* stochastic *part of the second-order requirement,* *while the bias part survives untouched (Remark 5.9). For a learner whose error is predominantly systematic, $D^2\Theta[e_A, e_B] \approx D^2\Theta[b, b] \approx D^2\Theta[\Delta, \Delta]$ and there is no gain at all. Note that the regular-DML rate $\|\hat{\gamma}^j - \gamma_0\|_\infty = o_p(n^{-1/4})$ is* not *imposed: irregularity already relaxes the second-order tolerance from $n^{-1/2}$ to $\sqrt{K_n^{1/d}/n}$, and the quadratic correction relaxes it again. The statement is conditional on (iii)–(iv) in the same sense as Theorem 4.6; see Appendix B, §B.4.*

Remark 5.9 (What condition (iv) hides)**.** *Condition (iv) is not innocuous, and it is where the limits of* any *split-sample device show up. Decomposing $e_j = b_j + u_j$ into a systematic and a mean-zero part, independence of the two folds gives $\mathbb{E}\, D^2\Theta[e_A, e_B] = D^2\Theta[b_A, b_B]$: the split-sample construction removes own-half* stochastic *diagonals, never the* bias product*. For sieve LS first stages with undersmoothing this is $O(K_n^{-(2s-1)/d}) = o(r_n^*)$ (Table 1). For a generic machine-learning first stage with a systematic bias — a tree ensemble's CATE attenuation, for instance, measured at* 13% *in Section 6.4 — $b_A \approx b_B \approx b$ is* not *small, and $D^2\Theta[b, b]$ is a genuine first-order-relevant term that neither the quadratic nor the Riesz correction touches. This is the formal counterpart of the Monte Carlo finding that boosting on the rough truth is bias-limited at* $+0.045$ *regardless of what corrections are applied. No debiasing scheme in this paper repairs a biased learner; they repair the* noise-induced *curvature of an otherwise adequate one.*

Proof sketch. Combine (46), (47) and (48) (derived in Appendix B, §B.3) with the conditional mean-zero property of the fold-C score. The leading term is $\frac{1}{3}\sum_\pi \frac{3}{n}\sum_{i \in C_\pi} v_{K_n}^* \epsilon_i = \frac{1}{n}\sum_i v_{K_n}^* \epsilon_i$, whose CLT is Assumption 5.1(g); the remaining terms are (iii), (iv), the representer's own approximation and estimation error (Assumption 5.1(d),(d′),(e)), and the out-of-span remainder R_5. □

Theorem 5.8 answers, in the two-threshold setting, the question left open in the single-threshold draft (its Remark 7) — but the answer is more delicate than v1–v2 claimed. The cross-fitted influence-function correction and the diagonal quadratic correction are compatible only if the first is taken with respect to the *composite* functional Ψ; correcting Θ fold-by-fold and then adding the quadratic term does not compose, because the one-step correction is itself second-order accurate only at the point where its representer is evaluated. Two further cautions. Condition (i) *fails* for tree-ensemble first stages (Remark 4.4), so Theorem 5.8 does not cover the GBR-based DD cells of Section 6. And those cells implement (45), not (49): see Section 6.5, item (L7).

6 Monte Carlo Evidence Calibrated to the Graduation Experiment

6.1 Design

Following the calibration philosophy of Chen and Ritzwoller (2023) — whose generative model is itself fit to the Banerjee et al. (2015) "graduation" RCT in Pakistan ($n = 854$; 2-year outcomes as surrogates S, 3-year consumption as Y) — we evaluate all estimators on three data-generating processes with *known* truth:

KRR a kernel-ridge exact-truth oracle: per-arm conditional means of (S, Y) fit to the graduation data by kernel ridge regression are taken as the *true* $\mu_{o,0}(\cdot, w)$; covariates are drawn from a Gaussian-KDE fit of the empirical covariate distribution ($d = 2$), truncated to $[-3, 3]^2$; residual noise is calibrated to the data with within-unit (S, Y) coupling ($\rho_{SY} \neq 0$). The truth $\theta_0 = 0.161$ and all margin/corner geometry are computable exactly. (See Section 6.5, items (L1)–(L3), for two implementation defects that displace this "exact" truth.)

WGAN the Chen and Ritzwoller (2023) three-GAN cascade $X \to S \mid X, W \to Y \mid S, X, W$ (the `ds-wgan` design of Athey et al., 2021) trained on the same data; the fitted generators are frozen as the truth and CATEs computed by common-random-number Monte Carlo. ReLU generators make the true contrast surfaces piecewise linear — effectively $s \approx 1$ — so this design sits *below* the smoothness threshold of every theorem in this paper and is a stress test, not a corroboration. We use two variants: the *scalar-surrogate* cascade, whose geometry has both margins binding ($\theta_0 = 0.194$), and the *full-surrogate* cascade at $d = 2$, which is *degenerate* — conditioning Y on the full 21-dimensional surrogate vector flattens τ_S and pushes $\tau_Y > 0$ everywhere, so $\theta_0 = 0$ with empty margins, violating Assumption 1.5(a) and (c). The degenerate arm is retained deliberately: it shows how the boundary-band inference behaves when the estimand sits on the boundary of the parameter space and no margin exists.

Affine a Gaussian–affine design with closed-form orthant truth, as a specification check of the numerical pipeline.

For each DGP we draw experiments of size $n \in \{1000, 2000, 4000, 8000\}$ (500 replications each) and compare, at nominal 95%: (i) the sieve plug-in $\hat{\Theta}$; (ii) margins-only SS $\hat{\Theta}_{SS}^{\mathrm{mar}}$; (iii) corner-aware SS $\hat{\Theta}_{SS}$ (32); (iv) the 2-fold cross-fitted GBR plug-in; (v) (iv) + projected Riesz correction; (vi) the doubly debiased $\hat{\Theta}_{DD}$ (49). A separate decomposition experiment

(Table 3) then varies the DML construction one axis at a time — fold count, first-stage learner (boosting, random forest, smooth random-feature ridge), and representer basis — to isolate the source of the machine-learning undercoverage. All are studentized by the two-band sieve variance (15).[1] Table 2 reports bias, MC standard deviation, RMSE, the mean-SE/MC-SD ratio, empirical coverage, and mean CI length; the estimated corner correction is examined separately in Section 6.2. With 500 replications the Monte Carlo standard error of a coverage figure near 0.95 is about 0.010, and near 0.90 about 0.013; differences smaller than roughly 0.03 should not be read as real.

Table 2: Calibrated Monte Carlo (500 replications per cell): bias, MC standard deviation, RMSE, mean-SE/MC-SD ratio, empirical 95% coverage, and mean CI length, by DGP, estimator, and sample size. All intervals are studentized by the two-band sieve-Riesz standard error (15) (cross-fitted residuals for the GBR family). Sieve dimension is fixed at $K = 25$ per arm for every n.

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
KRR (smooth truth), $\theta_0 = 0.161$							
Plug-in (sieve)	1000	+0.0103	0.0362	0.0376	1.11	0.962	0.158
	2000	+0.0052	0.0259	0.0264	1.06	0.958	0.108
	4000	+0.0037	0.0185	0.0188	1.00	0.954	0.073
	8000	+0.0031	0.0133	0.0136	0.94	0.948	0.049
SS margins-only	1000	-0.0058	0.0456	0.0459	0.88	0.882	0.158
	2000	-0.0050	0.0294	0.0298	0.94	0.918	0.108
	4000	-0.0021	0.0204	0.0205	0.91	0.932	0.073
	8000	-0.0005	0.0135	0.0135	0.93	0.926	0.049
SS corner-aware	1000	+0.0030	0.0446	0.0447	0.90	0.894	0.158
	2000	-0.0005	0.0293	0.0293	0.94	0.932	0.108
	4000	+0.0004	0.0204	0.0204	0.91	0.934	0.073
	8000	+0.0007	0.0134	0.0134	0.93	0.930	0.049
Cross-fit GBR	1000	+0.0666	0.0316	0.0737	1.41	0.680	0.175

[1]Implementation details: the sieve dimension is held *fixed* across n at $5 \times 5 = 25$ tensor B-spline basis functions per arm (two knot segments per coordinate), so the design varies n at fixed K rather than along $K_n \asymp n^{d/(2s+1)}$; see Section 6.5, item (L4). Bandwidths are $\varepsilon = \delta \cdot \widehat{\mathrm{sd}}(\hat\tau)$ with a trimmed standard deviation and $\delta = 0.08$ (sieve family) or 0.10 (ML family); if fewer than five evaluation points fall in a band it is widened by factors of 1.5 (at most six times) — an adaptivity that is active mainly on the degenerate arm, where the margins are empty. Ties are broken as $\mathbb{1}\{\hat\tau_S \geq 0\}$ and harm as the strict $\hat\tau_Y < 0$; the difference is measure-zero under Assumption 1.5.

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
	2000	+0.0444	0.0238	0.0504	1.06	0.578	0.099
	4000	+0.0239	0.0189	0.0304	0.87	0.666	0.064
	8000	+0.0088	0.0142	0.0167	0.82	0.820	0.046
+ Riesz corr.	1000	+0.0562	0.0545	0.0783	0.82	0.704	0.175
	2000	+0.0260	0.0268	0.0373	0.94	0.784	0.099
	4000	+0.0145	0.0196	0.0244	0.83	0.808	0.064
	8000	+0.0045	0.0141	0.0148	0.83	0.882	0.046
+ quad. (DD)	1000	+0.0488	0.0653	0.0815	0.68	0.706	0.175
	2000	+0.0061	0.0361	0.0366	0.70	0.830	0.099
	4000	-0.0046	0.0273	0.0277	0.60	0.752	0.064
	8000	-0.0109	0.0195	0.0223	0.60	0.690	0.046
WGAN scalar (rough truth), $\theta_0 = 0.194$							
Plug-in (sieve)	1000	+0.0645	0.1008	0.1195	1.34	0.956	0.527
	2000	+0.0592	0.0884	0.1063	1.15	0.928	0.398
	4000	+0.0470	0.0618	0.0776	1.23	0.936	0.298
	8000	+0.0317	0.0453	0.0552	1.13	0.952	0.200
SS margins-only	1000	+0.0435	0.1396	0.1461	0.96	0.886	0.527
	2000	+0.0373	0.1236	0.1290	0.82	0.866	0.398
	4000	+0.0275	0.0782	0.0828	0.97	0.918	0.298
	8000	+0.0152	0.0537	0.0557	0.95	0.940	0.200
SS corner-aware	1000	+0.0655	0.1370	0.1518	0.98	0.876	0.527
	2000	+0.0519	0.1213	0.1318	0.84	0.868	0.398
	4000	+0.0363	0.0780	0.0859	0.98	0.910	0.298
	8000	+0.0206	0.0535	0.0573	0.96	0.930	0.200
Cross-fit GBR	1000	+0.0452	0.0434	0.0626	2.17	0.960	0.369
	2000	+0.0463	0.0359	0.0585	1.18	0.840	0.166
	4000	+0.0469	0.0318	0.0567	1.03	0.726	0.128
	8000	+0.0415	0.0276	0.0499	0.97	0.656	0.105
+ Riesz corr.	1000	+0.0492	0.1242	0.1335	0.76	0.910	0.369
	2000	+0.0411	0.0417	0.0585	1.02	0.840	0.166
	4000	+0.0456	0.0335	0.0566	0.98	0.718	0.128
	8000	+0.0416	0.0280	0.0501	0.95	0.632	0.105
+ quad. (DD)	1000	+0.0609	0.1413	0.1537	0.67	0.772	0.369
	2000	+0.0446	0.0703	0.0832	0.60	0.656	0.166
	4000	+0.0336	0.0570	0.0661	0.57	0.660	0.128
	8000	+0.0209	0.0446	0.0492	0.60	0.702	0.105

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
WGAN full-surrogate (degenerate), $\theta_0 = 0.000$							
Plug-in (sieve)	1000	+0.0604	0.0432	0.0742	1.60	0.922	0.271
	2000	+0.0329	0.0243	0.0409	1.52	0.940	0.144
	4000	+0.0173	0.0113	0.0207	1.73	0.936	0.076
	8000	+0.0085	0.0061	0.0105	1.73	0.930	0.042
SS margins-only	1000	+0.0140	0.0669	0.0683	1.03	0.866	0.271
	2000	+0.0019	0.0362	0.0362	1.02	0.894	0.144
	4000	+0.0011	0.0188	0.0188	1.04	0.886	0.076
	8000	-0.0002	0.0092	0.0092	1.15	0.874	0.042
SS corner-aware	1000	+0.0314	0.0643	0.0715	1.07	0.888	0.271
	2000	+0.0101	0.0347	0.0361	1.06	0.906	0.144
	4000	+0.0050	0.0183	0.0190	1.06	0.898	0.076
	8000	+0.0018	0.0091	0.0093	1.16	0.898	0.042
Cross-fit GBR	1000	+0.1742	0.0357	0.1778	1.94	0.126	0.271
	2000	+0.1608	0.0300	0.1636	1.17	0.008	0.138
	4000	+0.1341	0.0231	0.1360	1.02	0.000	0.092
	8000	+0.0962	0.0175	0.0977	0.92	0.000	0.063
+ Riesz corr.	1000	+0.1732	0.1099	0.2051	0.63	0.146	0.271
	2000	+0.1598	0.0361	0.1638	0.97	0.026	0.138
	4000	+0.1342	0.0241	0.1363	0.98	0.000	0.092
	8000	+0.0966	0.0176	0.0982	0.91	0.000	0.063
+ quad. (DD)	1000	+0.1184	0.1203	0.1687	0.57	0.422	0.271
	2000	+0.0858	0.0523	0.1005	0.67	0.376	0.138
	4000	+0.0461	0.0353	0.0580	0.67	0.508	0.092
	8000	+0.0071	0.0200	0.0212	0.80	0.866	0.063
Affine (exact truth), $\theta_0 = 0.209$							
Plug-in (sieve)	1000	+0.0042	0.0546	0.0547	0.97	0.916	0.208
	2000	+0.0017	0.0339	0.0339	1.07	0.958	0.142
	4000	+0.0012	0.0249	0.0249	1.00	0.940	0.098
	8000	-0.0013	0.0171	0.0172	1.00	0.954	0.067
SS margins-only	1000	+0.0023	0.0680	0.0680	0.78	0.858	0.208
	2000	+0.0007	0.0392	0.0391	0.93	0.934	0.142
	4000	+0.0003	0.0267	0.0267	0.93	0.934	0.098
	8000	-0.0022	0.0178	0.0179	0.96	0.930	0.067
SS corner-aware	1000	+0.0025	0.0681	0.0680	0.78	0.860	0.208
	2000	+0.0005	0.0393	0.0393	0.92	0.928	0.142

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
	4000	+0.0003	0.0268	0.0268	0.93	0.932	0.098
	8000	-0.0022	0.0178	0.0179	0.96	0.932	0.067
Cross-fit GBR	1000	+0.0277	0.0378	0.0468	1.00	0.862	0.148
	2000	+0.0161	0.0284	0.0326	0.94	0.878	0.105
	4000	+0.0097	0.0238	0.0257	0.85	0.886	0.079
	8000	+0.0036	0.0174	0.0178	0.86	0.900	0.059
+ Riesz corr.	1000	+0.0255	0.0416	0.0487	0.91	0.870	0.148
	2000	+0.0140	0.0291	0.0322	0.92	0.896	0.105
	4000	+0.0077	0.0241	0.0253	0.84	0.876	0.079
	8000	+0.0022	0.0182	0.0184	0.83	0.886	0.059
+ quad. (DD)	1000	+0.0032	0.0604	0.0604	0.63	0.764	0.148
	2000	-0.0024	0.0441	0.0441	0.60	0.778	0.105
	4000	-0.0014	0.0325	0.0325	0.62	0.780	0.079
	8000	-0.0036	0.0241	0.0243	0.63	0.778	0.059

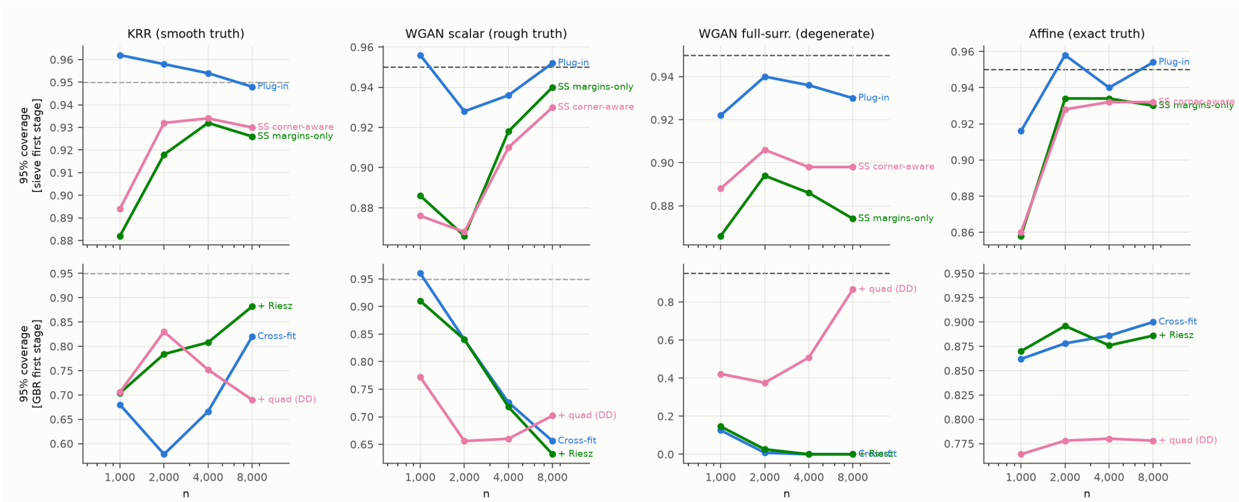

Figure 1: Empirical 95% coverage by sample size, DGP (columns), and first-stage family (rows). Dashed line: nominal 0.95.

6.2 Corner anatomy

Because the KRR oracle's contrast surfaces are known in closed form, the corner set and its geometry are computable: the zero curves of τ_S and τ_Y cross at ten points, with gradient norms $\|\nabla\tau_S\| \in [10.3, 135.8]$, $\|\nabla\tau_Y\| \in [13.9, 56.4]$, $|\cos\omega| \leq 0.90$ (transversality holds:

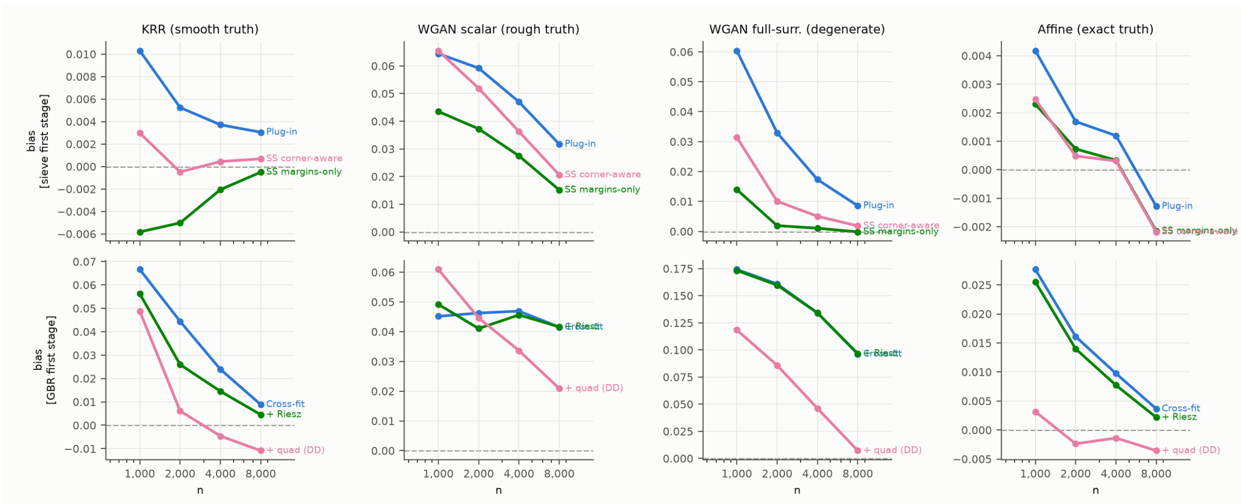

Figure 2: Monte Carlo bias by sample size, DGP, and estimator. Dashed line: zero.

Table 3: Decomposition of the machine-learning undercoverage on the KRR design ($n = 4000$, 300 replications). "Plug-in bias" is the uncorrected cross-fit bias; "bias", "SD", "SE/SD", and "coverage" refer to the Riesz-corrected estimator, studentized by (15). The tensor-sieve plug-in is the reference. Varying folds, learner, and representer basis one at a time isolates each mechanism of Remark 5.5. MC standard error of a coverage figure is about 0.013 at 300 replications.

Configuration	Bias	Plug-in bias	SD	SE/SD	Coverage
tensor-sieve plug-in	+0.0034	+0.0034	0.0178	1.03	0.943
gbr K=2 riesz=sieve	+0.0216	+0.0317	0.0186	0.96	0.760
gbr K=5 riesz=sieve	+0.0106	+0.0160	0.0191	0.94	0.903
gbr K=10 riesz=sieve	+0.0079	+0.0127	0.0196	0.92	0.913
rf K=5 riesz=sieve	+0.0119	+0.0453	0.0195	1.01	0.907
krr K=5 riesz=sieve	+0.0038	+0.0039	0.0184	0.90	0.917
gbr K=5 riesz=rf	+0.0078	+0.0160	0.0203	0.96	0.937
krr K=5 riesz=rf	+0.0047	+0.0039	0.0181	0.98	0.937
gbr(high-cap) K=5 riesz=sieve	+0.0439	+0.0461	0.0131	1.00	0.077
gbr K=5 no correction	+0.0160	+0.0160	0.0189	0.94	0.857

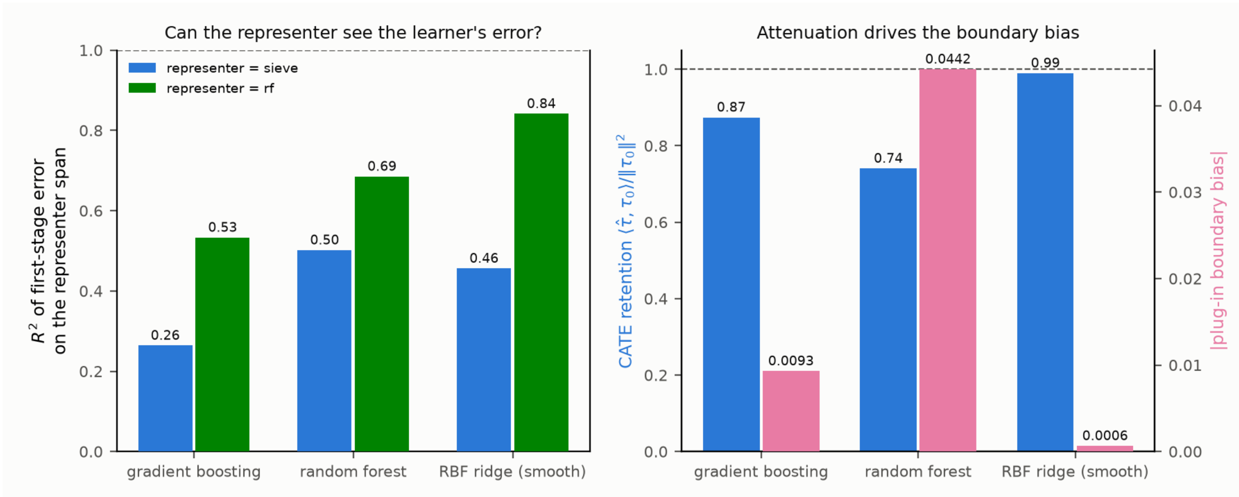

Figure 3: Why boosting-based DML undercovers. *Left*: the fraction (R^2) of the first-stage error that the sieve-Riesz representer can span in global L^2, measured against the oracle arm means; a boosted-tree error is largely invisible to a smooth representer, a random-feature ridge's error is not. *Right*: tree learners attenuate the CATE (no-intercept projection coefficient of fitted on true, $\hat{\tau}'\tau/\tau'\tau < 1$), which displaces the decision boundary and biases the plug-in; the smooth learner barely attenuates and is nearly unbiased. Both panels average five simulated datasets and are indicative rather than precise.

Table 4: Corner anatomy on the KRR-calibrated oracle (300 reps, fixed sieve dimension). "Total" is the implied plug-in corner bias $\sum_{x^*} \mathbb{E}[\Delta_g \Delta_h]\, w/(\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|)$ in θ units; "SS estimate" is the mean corner cross correction $\widehat{\text{corner}}$ of (35), which by split-sample independence targets the *stochastic* (K/n) part of the diagonal; "remainder" is their difference, the bias-product ($K^{-2s/d}$) part. Sign convention: the anatomy script measures $\mathbb{E}[\Delta\hat{\tau}_S\,\Delta\hat{\tau}_Y]$ (positive here, as $\rho_{SY} > 0$); the table reports the corner term in θ units, where the harm-share convention $h_0 = -\tau_Y$ flips its sign. $\overline{\text{corr}}$ averages $\text{Corr}(\Delta_g(x^*), \Delta_h(x^*))$ over the ten corners. The column implements the corner *cross* term $\mathcal{B}_{\mathcal{C}}^{\times}$ of Proposition 4.10 — the first term of the bracket in (27) — which by (37) is exactly the quantity that margins-only debiasing leaves behind and that $\widehat{\text{corner}}$ targets, so the comparison across columns is like-for-like. It is *not* the corner's full contribution to the plug-in bias, which also contains the two $\cos\omega$-weighted diagonals of (27) (non-negligible here, since $|\cos\omega|$ reaches 0.90) but which margins-only debiasing already removes. The column also uses an unnormalized covariate density (Section 6.5, item (L2)).

n	Total corner bias	SS estimate	Remainder	$\overline{\text{corr}}(\Delta_g, \Delta_h)$	ρ_{SY}
2000	-0.0080	-0.0044	-0.0036	0.50	0.524
4000	-0.0048	-0.0022	-0.0025	0.46	0.524
8000	-0.0038	-0.0012	-0.0026	0.45	0.524

$|\sin\omega| \geq 0.45$), and within-unit residual correlation $\rho_{SY} = 0.524$. Table 4 traces the corner mechanism of Section 3.2 across 300 replications per sample size (sieve first stage, fixed K).

Three predictions of Section 3.2 are consistent with the table. *(i)* The pointwise correlation of the two surface errors at the corner tracks the within-unit residual correlation (0.45–0.50 against $\rho_{SY} = 0.524$), which is the mechanism behind the V_{gh} term in (27). *(ii)* The stochastic part of the corner diagonal halves as n doubles at fixed K ($-0.0044 \to -0.0022 \to -0.0012$) — the K/n scaling — and is tracked by the SS corner correction, which by construction cancels bias parts across the two independent halves. *(iii)* The remainder is roughly constant in n at fixed K (-0.0036 at $n = 2000$, then ≈ -0.0025), as befits the $K^{-2s/d}$ bias product, which is controlled by undersmoothing K rather than by debiasing — and which, per Proposition 4.10, the SS statistic is not supposed to capture. The measured corner bias (0.4–0.8 percentage points of a $\theta_0 = 0.161$ target) is material relative to the plug-in standard error, and its sign matches the Monte Carlo finding that the corner-aware and margins-only SS estimates differ by $+0.004$ to $+0.005$ at $n = 2000$. This validates the cross-corner component of (27) — the component that Proposition 4.10 says is at stake in the margins-only comparison. The two $\cos\omega$ components of (27) are not tested here, because both debiasing variants remove them; testing them would require comparing the *plug-in* against a corner-corrected plug-in.

6.3 Direct test of the cross-half off-diagonal order

Table 2 contains, without any further computation, a parameter-free test of the off-diagonal order derived in Appendix B, §B.2 — the one input to Theorem 4.6(ii)'s window (39) that is otherwise untested.

The SS estimator's error differs from the plug-in's by removing the stochastic diagonal and adding the mean-zero cross-half term $\frac{1}{2}D^2\Theta[u^A, u^B]$. By §B.2 that term has variance $\asymp n^{-2}K_n^{(d+3)/d}$, against a score variance $\sigma_n^2/n \asymp K_n^{1/d}/n$. Since the two estimators use the same observations and the same standard error, this predicts

$$\frac{\mathrm{SD}(\hat{\Theta}_{SS})}{\mathrm{SD}(\hat{\Theta})} = \sqrt{1 + c\,\frac{K_n^{(d+2)/d}}{n}\,(1 + o(1))}, \qquad c \text{ constant in } n, \tag{50}$$

which at the design's fixed $K = 25$, $d = 2$ is $\sqrt{1 + 625c/n}$. Writing R_n for the observed ratio $\mathrm{SD}_{SS}/\mathrm{SD}_{\text{plug-in}}$, the implied $c = n(R_n^2 - 1)/K^2$ should be flat in n:

Design	$n = 1000$	2000	4000	8000	spread
Affine (exact truth; assumptions hold)	0.89	1.10	1.01	1.07	1.2×
KRR (smooth)	0.83	0.90	1.38	0.19	7.2×
WGAN scalar ($s \approx 1$; violates)	1.36	2.83	3.80	5.05	3.7×
WGAN degenerate (empty margins; violates)	1.94	3.33	10.39	15.69	8.1×

On the affine design — the only one with an exactly closed-form truth and all maintained assumptions satisfied — c is flat within a factor 1.2 and centered on 1, across a factor of eight in n, with no fitted parameter beyond the $\asymp$ constant itself. On KRR the first three cells agree; the $n = 8000$ cell is uninformative, since there $(\mathrm{SD}_{SS}/\mathrm{SD}_{\text{plug-in}})^2 - 1 = 0.016$ is of the order of the Monte Carlo error of a ratio of two standard deviations estimated from 500 replications (we have not computed that error, which is smaller than an independent-samples calculation would give, since the two use *common* replications). The two designs that violate the maintained assumptions show c increasing monotonically by factors of 3.7 and 8.1, flatly inconsistent with (50). The test therefore discriminates rather than merely fits.

Two corollaries. First, this *completely* explains the SS standard error's calibration. Because the implementation computes one standard error from the full-sample fit and applies it to both estimators, SE/SD for SS must equal (SE/SD for the plug-in) divided by the SD ratio, as an identity; and indeed the reported values agree to within 0.003–0.008 on all four designs and all four sample sizes. The SS SE/SD deficit of 0.78–0.96 is thus *entirely* the variance of the quadratic correction, which the first-order standard error omits by construction — not a failure of the two-band variance formula, and not the corner.

Second, the finite-sample deficit is the visible shadow of a condition in Theorem 4.6(ii). The inflation factor $K_n^{(d+2)/d}/n$ vanishes precisely when $a < d/(d+2)$, which is the *upper edge of the window* (39). At the design's fixed $K = 25$ the factor is $0.63, 0.31, 0.16, 0.08$ — large at $n = 1000$, still 8% at $n = 8000$ — and a growing-K_n implementation inside (39) is exactly what would drive it away. A standard error for $\hat{\Theta}_{SS}$ that included the correction's own variance would fix the calibration at fixed K; we do not currently have one.

6.4 Results

Table 2 and Figures 1–2 report all cells; Section 6.5 states which conclusions they can and cannot support.

(1) Smooth truth (KRR): the plug-in is already good; corner-aware SS removes what bias there is. The plug-in covers at 0.948–0.962 at all n with a small decaying bias ($+0.010 \to +0.003$) and a well-calibrated standard error (SE/SD 0.94–1.11, tightening toward one with n). SS debiasing removes the bias essentially completely ($+0.003 \to +0.0007$; a 3.5-fold bias

reduction at $n = 1000$ and a tenfold reduction at $n = 2000$), at a variance cost that is 23% of the standard deviation at $n = 1000$ and falls to 10% at $n = 4000$ and about 1% at $n = 8000$ — Section 6.3 shows this is the cross-half term $\frac{1}{2}D^2\Theta[u^A, u^B]$, decaying as $K_n^{(d+2)/d}/n$ predicts, and that it accounts for the whole of the SS coverage shortfall. The *margins-only* correction overshoots to negative bias (-0.006 to -0.0005) at every n: it removes the (positive) margin diagonals — and, per (37), the two $\cos\omega$ corner diagonals as well — but not the (negative) corner *cross* term. The gap between the corner-aware and margins-only estimates equals the estimated corner correction ($\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\mathrm{mar}} - \widehat{\mathrm{corner}}$) and matches the anatomy of Table 4 to the fourth decimal ($+0.0045$ at $n = 2000$ against minus the anatomy's SS corner estimate of -0.0044). The raw corner cross form $q_{\mathrm{full}} - q_g - q_h$ (eight times the corner correction) halves as n doubles ($-0.070 \to -0.036 \to -0.020 \to -0.0095$) at fixed K, the K/n scaling of (27). Coverage of the SS intervals is 0.894–0.934 on this design: an improvement in centering bought at a 2–5 point coverage cost relative to the plug-in, which we do not describe as honest coverage.

(2) Rough truth (scalar WGAN, $s \approx 1$): debiasing accelerates the bias decay. This design violates the smoothness hypothesis of every theorem above ($s \approx 1 < (d{+}1)/2 = 1.5$), so it is a robustness probe. The plug-in bias is large and decays at only $\approx n^{-0.34}$ ($+0.064 \to +0.032$), and its near-nominal coverage rests on a conservative SE (SE/SD 1.1–1.3). SS debiasing accelerates the empirical decay to $\approx n^{-0.55}$ ($+0.066 \to +0.021$) with SE/SD 0.83–0.98 and coverage $0.87 \to 0.93$. Because K is fixed across n here, these exponents describe the finite-sample behavior of the estimators at $K = 25$; they are not estimates of the asymptotic rate, which would require K_n to grow (Section 6.5, item (L4)). On this rough truth the margins-only variant is slightly better centered than the corner-aware one — with $s \approx 1$ the corner correction is itself noisily estimated — a useful caution: corner-awareness is a bias-*decomposition* necessity, but its sampled version adds noise when the surfaces are rough.

(3) Machine-learning first stages: the projected Riesz correction works where the learner is adequate; the quadratic correction is not a free lunch. On the smooth truth the GBR bias falls monotonically along the ladder cross-fit $\to$ +Riesz $\to$ +quadratic at every n (at $n = 4000$: $+0.024 \to +0.015 \to -0.005$), consistent with the two corrections targeting distinct error terms. But the composition has two finite-sample costs: the SS-on-GBR quadratic correction inflates the dispersion (by a factor of two at $n = 1000$, 1.4 thereafter), and the first-order SE does not account for it (SE/SD ≈ 0.6), so DD intervals undercover (0.69–0.83); and at $n = 8000$ on KRR the quadratic correction overshoots (-0.011). *This ladder must now be read with item (L7) of Section 6.5 in hand*: the DD column implements the superseded construction (45), whose second-order bias $-\frac{1}{2}\{D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]\}$ predicts exactly this over-correction and this dispersion inflation, on smooth and rough designs alike. v2

attributed the overshoot to Remark 4.4 — (32) being a fourth-order finite difference of a step function for tree first stages — which is a real effect but cannot explain the same pattern on the affine design. On the rough truth GBR is bias-limited (+0.045 nearly flat in n; coverage falls to 0.64): no first-order correction can rescue a nuisance that misses the kinks. The degenerate arm is the extreme case: GBR manufactures a spurious harm share of +0.17 (coverage 0), and there the quadratic correction *does* rescue it (+0.007, coverage 0.87 at $n = 8000$).

(4) The two-band SE is well calibrated for the sieve family. The variance estimator (15) includes the empirical-measure term $\hat{\varsigma}^2/n = \hat{\theta}(1-\hat{\theta})/n$ — the sampling error of the average given the surfaces, the analogue of the $\mathrm{Var}([h_0]_+)$ term the single-threshold unknown-density welfare theorem adds — which is 6–12% of the SE at these sample sizes and, if omitted, produces a spurious SE/SD of 0.87–0.95. With it included, SE/SD is 0.90–1.11 for plug-in and SS on the KRR and affine designs, and the studentized intervals reach 0.93–0.95 by $n = 8000$; on the degenerate arm ($\theta_0 = 0$, empty margins) the sieve family degrades gracefully.

(5) Machine-learning first stages: why the sieve-DML interval undercovers, and when it does not. The cross-fitted GBR estimators cover only 0.58–0.88 on the smooth design, in sharp contrast to the sieve plug-in on the *same samples* — so the shortfall is a property of the first stage, not the DGP or the variance formula. The diagnosis is Remark 5.5. Measured against the oracle arm means, the R^2 of the boosted first-stage error on the tensor B-spline representer span is 0.21–0.34 across the four arm regressions (mean 0.26); the projected correction removes 31% of the plug-in's boundary bias. The direct variance test reported in Remark 5.5 shows that the remaining 69% is *bias*, not an independent added variance: the leading explanation is that the correction cannot see the part of the learner's error that lives on the margins, so it leaves the boundary displaced. Three levers move coverage: *(i) folds* — $K = 2 \to 5 \to 10$ lifts GBR coverage $0.76 \to 0.90 \to 0.91$ by shrinking the own-observation bias trained on $(K-1)/K$ of the sample; *(ii) representer basis* — matching a 400-feature random-feature representer to the boosted learner raises the projected R^2 from 0.26 to 0.53 and coverage from 0.90 to 0.94; *(iii) learner smoothness* — a random-feature (RBF) ridge, whose error the representer *can* span ($R^2 = 0.84$), attenuates the CATE by only about 1% (against boosting's 13%) and has a plug-in bias of +0.0006 (against +0.016), so there is almost nothing left to correct and, with a matched representer, it too reaches 0.94. Higher-capacity boosting is counterproductive — it sharpens the axis-aligned staircase, driving coverage toward zero. The pattern reproduces the parent draft's own single-threshold Table 7. The residual 0.01–0.02 shortfall of even the best machine-learning cell below the sieve plug-in's coverage is within two Monte Carlo standard errors and should not be over-interpreted.

6.5 What the current Monte Carlo does and does not establish

The tables above are reproduced from v1 without change. Auditing the code against the theory turned up six implementation gaps. None of them affects Theorem 3.1 or the order accounting of Section 3.2, which are analytic; all of them bear on how much weight the simulation evidence can carry, and all are scheduled for repair before the next revision. We list them explicitly rather than leave them implicit.

(L1) The KRR "oracle truth" is displaced by uncentered residual resampling. The sampler draws outcome noise by resampling the fitted per-arm KRR residual pools, which are not centered (ridge shrinkage leaves arm-specific residual means). The generated conditional means therefore differ from the nominal KRR surfaces by a constant per arm, shifting the generated CATEs by $\Delta\tau_S = +0.126$ and $\Delta\tau_Y = +0.182$ (the latter includes the short→long coupling). The reported truth θ_0 moves from 0.16115 to 0.15998 — immaterial for the bias columns — but the ten corner locations, gradients, angles, and first-stage errors used in Table 4 are all computed against the *wrong* surfaces. Fix: center each arm's residual pool before resampling, or define the oracle truth to include the shifts.

(L2) The corner density is unnormalized. Covariates are drawn from the KDE conditional on $X \in [-3, 3]^2$ by rejection sampling, but the density accessor returns the untruncated KDE. The KDE mass of the box is 0.807, so the corner integrals in Table 4 should be scaled by $1/0.807 \approx 1.24$ before any comparison with (27). Together with the omission of the $\cos\omega$ terms noted in the table caption, the "Total" column is an incomplete implementation of the theory it is meant to check.

(L3) The KRR design violates Assumption 1.5(d). The KDE density is positive at the truncation boundary, and a direct scan of the four box edges finds twelve sign crossings of the two contrast surfaces (τ_S: three on $x_1 = -3$, two on $x_1 = +3$, three on $x_2 = -3$; τ_Y: two on $x_1 = -3$, two on $x_2 = +3$). The margins therefore terminate at the support edge, and by Remark 1.10 the correct second derivative for this DGP carries additional support-edge $\mathcal{H}^{d-2}$ terms that (18) omits. Fix: taper w near $\partial\mathcal{X}$, or shrink the box so that all margins close inside it.

(L4) The main Monte Carlo holds K fixed. Every sample size uses two knot segments per coordinate, i.e. $K = 25$ basis functions per arm at every n. The design therefore does not implement $K_n \asymp n^{d/(2s+1)}$, and the estimated log-rate slopes reported in Section 6.4(2) are not estimates of the asymptotic rate. What the design *does* deliver cleanly is the K/n scaling of the corner diagonal at fixed K (Table 4), which is the prediction Section 3.2 is actually testing.

(L5) The variance bandwidth does not satisfy Proposition 2.4.
The implementation sets $\varepsilon = \delta \cdot \widehat{\mathrm{sd}}(\hat{\tau})$ with δ fixed and then widens ε by factors of 1.5 until at least five observations enter the band. Since $\widehat{\mathrm{sd}}(\hat{\tau})$ converges to a positive constant, $\varepsilon_n \not\to 0$, contradicting Proposition 2.4 and Assumption 5.1(b). On the degenerate arm the widening manufactures a nonempty derivative band where the population margin is empty. Fix: $\varepsilon_n = \delta_n \cdot \widehat{\mathrm{sd}}(\hat{\tau})$ with $\delta_n \to 0$ at a rate satisfying $n\varepsilon_n/K_n \to \infty$.

(L6) The coded Riesz correction is not fold-specific.
Both the cross-fitted DML estimator and the DD estimator build a single full-sample Gram matrix and band representer from pooled out-of-fold predictions, whereas (42) and Theorem 5.3 require an off-fold representer for each fold. In the implementation, observation i's outcome can influence the global band through the model trained on i's own fold, which breaks the conditional mean-zero step used in Remark 5.5. Fix: build $\hat{v}^*_{-\kappa}$ inside the fold loop.

(L7) The DD rows implement the superseded construction (45).
Every "+ quad. (DD)" cell of Table 2 evaluates $\hat{\Theta}^{\mathrm{v2}}_{DD}$, which by Remark 5.6 carries a second-order bias $-\frac{1}{2}\{D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]\}$ that the corrected estimator of Definition 5.7 does not. These cells therefore say nothing about Theorem 5.8. They are, however, *consistent* with the diagnosis: since the uncorrected plug-in bias is positive on every design, a term $-\frac{1}{2}D^2\Theta[e, e]$ of the same sign as that bias predicts an over-correction into negative territory, which is what the smooth designs show (-0.0046 and -0.0109 at $n = 4000, 8000$ on KRR; -0.0024 to -0.0036 on Affine) together with the dispersion inflation (SE/SD ≈ 0.6 throughout). v2 attributed this over-correction to the non-smoothness of tree first stages (Remark 4.4); the structural defect is the better explanation, because it is present for the smooth (KRR, affine) designs as well, where the tree explanation does not apply. Fix: implement Definition 5.7 with three folds.

Taken together: Theorem 3.1 is verified analytically (Appendix C); the K/n order of the corner diagonal and the sign and magnitude of the corner correction are supported by Tables 4 and 2 at $d = 2$ and fixed K, subject to (L1)–(L3); the cross-half off-diagonal order $n^{-1}K_n^{(d+3)/(2d)}$ of Appendix B, §B.2 — which sets the upper edge of the SS window (39) — is confirmed by the parameter-free test of Section 6.3 on the designs that satisfy the assumptions and refuted on those that do not; the asymptotic rate claims of Theorems 2.2, 4.6, 5.3 and 5.8 are *not* tested by the present design, because K does not grow with n; the

DD cells test a construction we have since shown to be invalid, (L7); and the mechanism behind the machine-learning undercoverage is identified as residual boundary bias rather than unmodelled variance, by the direct variance test in Remark 5.5.

Practical guidance. We state it narrowly, because Section 6.5 limits what the experiments support and Section 5 limits what the theory establishes.

For sieve first stages, the corner-aware SS estimator (32) with the two-band SE is the recommendation: it is tuning-free and removes the second-order stochastic diagonal including the corner. In our designs its centering is clearly better than the plug-in's and its coverage is 2–5 points worse (0.894–0.934 on KRR), so it buys bias reduction at a coverage cost; we would not call its intervals honest. Undersmoothing relative to $K_n \asymp n^{d/(2s+1)}$ is required for the interval to be centered at all — window (39) — and the fixed-K Monte Carlo does not exercise this.

For a machine-learning first stage — needed when d is too large for a tensor sieve — prefer a *smooth* learner and *match the representer basis to the learner's own feature map*, so the projected correction is near-exact; avoid tree ensembles, whose piecewise-constant error is largest precisely on the margins the representer must span; and use at least five folds for the nuisance cross-fit. (This concerns the number of *nuisance* folds and does not conflict with the two-half construction of $\hat{\Theta}_{SS}$: there, the two halves furnish the independent replicates that the polarization identity (31) requires, and $K = 2$ is part of the definition of the quadratic correction, not a cross-fitting choice. The corrected DD estimator of Definition 5.7 needs a third fold on top of that.) When a tree first stage is unavoidable, pair it with a full-refit bootstrap rather than the analytic SE.

We do *not* currently recommend the doubly debiased estimator in applied work. The construction evaluated in Table 2 is invalid (Remark 5.6); the corrected construction of Definition 5.7 has not been implemented or simulated; and even for it, Theorem 5.8 is conditional on remainder bounds we have not established and excludes tree first stages outright.

The comparison of the corner-aware and margins-only corrections is a useful specification diagnostic in its own right: their gap estimates the corner cross term, whose magnitude flags whether the two-threshold structure is quantitatively binding.

A Derivation of the Second Derivative (Theorem 3.1)

Write $T_g(t) := \int_{\{g_t=0\}\cap\{h_t\geq 0\}} \varpi_t \, \mathrm{d}\mathcal{H}^{d-1}$ with weight $\varpi_t := u\, w/\left\|\nabla g_t\right\|$ (we reserve $\omega(x)$ for the transversality angle throughout), the first term of (10) along γ_t; the second term is symmetric. Let $\mathbf{X}(\cdot, t)$ be the normal diffeomorphism carrying $\mathcal{M} := \{g_0 = 0\}$ to $\{g_t = 0\}$

with velocity $\dot{\mathbf{X}} = -(u/\|\nabla g_0\|)\, n_g$, and pull the integral back to $\mathcal{M}$:

$$T_g(t) = \int_{\mathcal{M}} \mathbb{1}\{h_t(\mathbf{X}(x,t)) \geq 0\}\; \varpi_t(\mathbf{X}(x,t))\, J_{\mathbf{X}}(x,t)\, \mathrm{d}\mathcal{H}^{d-1}(x).$$

Differentiating at $t = 0$ yields four contributions. *(1) Integrand:* $\partial_t \varpi_t = -u\, w\, (n_g \cdot \nabla u)/\|\nabla g_0\|^2$. *(2) Transport of* ϖ *and area:* by the surface-transport formula (Delfour and Zolésio, 2001, Ch. 9), $\partial_t\{\varpi(\mathbf{X}) J_{\mathbf{X}}\} = -\big[(u/\|\nabla g_0\|)\, n_g \cdot \nabla \varpi + \varpi\, (u/\|\nabla g_0\|) H_g\big]$, the moving-level-set terms of Chen and Gao (2026, Lem. PathD–Level) with weight ϖ. *(3) Direct motion of the cut:* the indicator $\mathbb{1}\{h_t \geq 0\}$ evaluated on the (fixed, $t = 0$) manifold defines the region $\{h_0 + tv \geq 0\} \cap \mathcal{M}$; by the within-manifold Leibniz rule its derivative is an integral over the relative boundary $\mathcal{C}$ with normal velocity $v/\|\nabla_{\mathcal{M}} h_0\|$, where $\|\nabla_{\mathcal{M}} h_0\| = \|P_T \nabla h_0\| = \|\nabla h_0\|\, |\sin\omega|$, giving $+\int_{\mathcal{C}} \varpi_0\, v/(\|\nabla h_0\|\, |\sin\omega|)\, \mathrm{d}\mathcal{H}^{d-2}$ — the cross term. *(4) Induced motion of the cut:* the composition $h_0(\mathbf{X}(x,t)) = h_0(x) - t\,(u/\|\nabla g_0\|)\, n_g \cdot \nabla h_0 + o(t) = h_0(x) - t\, u\, \|\nabla h_0\| \cos\omega/\|\nabla g_0\| + o(t)$ perturbs the cut function by an effective $v_{\text{eff}} = -u\, \|\nabla h_0\| \cos\omega/\|\nabla g_0\|$; the same within-manifold Leibniz rule then contributes $-\int_{\mathcal{C}} \varpi_0\, u \cos\omega/(\|\nabla g_0\|\, |\sin\omega|)\, \mathrm{d}\mathcal{H}^{d-2}$ — the diagonal corner term of (19). Collecting (1)–(4), and adding the symmetric derivative of the $\mathcal{M}_h$-term (which contributes the mirror-image margin form, the second copy of the cross term, and the v^2 diagonal corner term), gives (18)–(19). Transversality bounds every corner integrand; twice-differentiability of g_0, h_0 and Assumption 1.4(d) bound the curvature and gradient terms. $\square$

B Orders of the Remainder Terms

The margin components of all remainder bounds are adaptations of the single-threshold arguments — Chen and Gao (2026, Thms. 5, 8 and the SS/LOO constructions) for the quadratic terms, and the extended single-threshold draft (sieve-DML theorem and its Assumption 5) for the cross-fitted first-order correction — applied margin by margin with the truncation indicators carried as bounded weights. This appendix records the two order calculations that are specific to the two-threshold geometry and that v1 got wrong.

B.1 Why the margin diagonal is $K_n^{1+1/d}/n$

Substituting the stochastic part (21) into Q_g and separating $i = j$ from $i \neq j$,

$$Q_g[u_g, u_g] = \underbrace{\frac{1}{n^2} \sum_i \epsilon_{S,i}^2\, Q_g[s_i, s_i]}_{T_{\text{diag}}} + \underbrace{\frac{1}{n^2} \sum_{i \neq j} \epsilon_{S,i} \epsilon_{S,j}\, Q_g[s_i, s_j]}_{T_{\text{off}}}.$$

The leading integrand of Q_g carries one gradient, so $\mathbb{E}\, Q_g[s_i, s_i]$ involves $\mathbb{E}[s_i(x)\, n_g{\cdot}\nabla s_i(x)] = n_g \cdot \{b(x)'G^{-1}\nabla b(x)\}$, whose modulus is $\lesssim K_n^{1+1/d}$ by (22), whence $\mathbb{E}\, T_{\text{diag}} = O(n^{-2} \cdot n \cdot K_n^{1+1/d}) = O(K_n^{1+1/d}/n)$. This is the *bound* stated by Chen and Gao (2026) for the single-threshold value functional, and it is the bound that Table 1 records. The one-sided form is deliberate: the quantity is signed, and Remark 3.4 shows that its surface integral can be an order smaller. The curvature part of Q_g, which carries no gradient, contributes $O(K_n/n)$.

The corner forms in (18)–(19) carry *no* gradient of u or v. Their diagonals therefore involve $\mathbb{E}[s_i(x)^2] = b(x)'G^{-1}b(x) = \mathcal{V}(x) \asymp K_n$ and are $O(K_n/n)$: bounded by $K_n^{-1/d}$ times the margin bound. v1's assertion that the corner diagonal is "of the same order K_n/n as the margin diagonals" conflated a bound with an exact order.

Two asymmetries between the two statements should be recorded. *(i)* The corner order is exact: $\mathcal{V} > 0$ and enters undifferentiated, so integrating it over $\mathcal{C}$ against a weight of constant sign cannot cancel. *(ii)* The margin order is a bound only. $\mathbb{E}[s_i\, \partial_n s_i] = \frac{1}{2}\partial_n \mathcal{V}$ under homoskedasticity, and $\mathcal{V}$ oscillates at the mesh scale with amplitude $\asymp K_n$, so while $\|\nabla \mathcal{V}\|_\infty \asymp K_n^{1+1/d}$ pointwise, the surface integral in (29) can cancel. Expanding $\mathcal{V} = \bar{\mathcal{V}} + \tilde{\mathcal{V}}$ with $\tilde{\mathcal{V}}$ mesh-periodic and mean zero, and writing $\tilde{\mathcal{V}}(x) = \sum_{j \neq 0} c_j e^{2\pi i j \varkappa^{-1} x}$, the integral of $\partial_n \mathcal{V}$ along a margin of slope α (in $d = 2$) picks up a non-decaying contribution only from pairs (j, l) with $j + l\alpha \approx 0$, i.e. only if α is rational with small denominator; for cubic splines $|c_j| = O(j^{-4})$, so even moderate denominators are heavily suppressed. Appendix C, item 5, confirms this: $\alpha \in \{0, 1\}$ give $\asymp K_n^{1+1/d}$, $\alpha \in \{1/2, 8/5\}$ and irrational α give $\asymp K_n$. Consequently the margin bound is attained for mesh-resonant margins — in particular for $d = 1$, where $\mathcal{M}_g$ is a point and no averaging occurs, so the single-threshold rate of Chen and Gao (2026) is unaffected — but not necessarily otherwise.

B.2 Off-diagonal (cross-half) orders

T_{off} is a degenerate second-order U-statistic. Writing $a_i := G^{-1}b(X_i)$, every quadratic form in (18) evaluates to $a_i'Ma_j$ for a symmetric matrix M built from the relevant boundary integral, so $\text{Var}(T_{\text{off}}) \asymp n^{-2}\,\text{tr}\{(MG^{-1})^2\}$. For tensor B-splines with mesh $\varkappa = K_n^{-1/d}$, a normalized basis function has height $\asymp \varkappa^{-d/2}$ on a cell of volume $\varkappa^d$, so for a p-dimensional integration set $\mathcal{S}$ and ℓ gradient factors,

$$\#\{k : \text{supp}(b_k) \cap \mathcal{S} \neq \emptyset\} \asymp \varkappa^{-p}, \qquad M_{kk} \asymp \varkappa^{-d}\varkappa^{p}\varkappa^{-\ell} = \varkappa^{p-d-\ell},$$

and therefore $\text{tr}(M^2) \asymp \varkappa^{-p}\varkappa^{2(p-d-\ell)} = \varkappa^{p-2d-2\ell} = K_n^{(2d+2\ell-p)/d}$. Substituting:

Term	(p, ℓ)	$\mathrm{tr}(M^2)$	T_{off}
margin, gradient part	$(d-1, 1)$	$K_n^{(d+3)/d}$	$n^{-1}K_n^{(d+3)/(2d)}$
corner	$(d-2, 0)$	$K_n^{(d+2)/d}$	$n^{-1}K_n^{(d+2)/(2d)}$
margin, curvature part	$(d-1, 0)$	$K_n^{(d+1)/d}$	$n^{-1}K_n^{(d+1)/(2d)}$

The first line reproduces the $n^{-1}\sqrt{K_n^{(d+3)/d}}$ bound of Chen and Gao (2026), which is a useful check on the calculus. Two consequences. First, all three are $O(r_n^*)$ at $K_n \asymp n^{d/(2s+1)}$ under $s \geq (d+1)/2$ (the corner needs only $s \geq d/2$), so the off-diagonals never bind before the terms in Table 1's third block do. Second, and correcting v1: it is *not* true in general that a lower-dimensional integration set gives a smaller cross-half term. Reducing p by one *raises* $\mathrm{tr}(M^2)$ by $K_n^{1/d}$, because there is less averaging, not more. Comparing like with like, the codimension-two corner has a *larger* generic off-diagonal than a hypersurface with the same integrand. The corner off-diagonal ends up smaller than the leading margin off-diagonal only because the margin form carries a gradient factor ($\ell = 1$) that costs $K_n^{2/d}$ in $\mathrm{tr}(M^2)$, which more than offsets the corner's $K_n^{1/d}$ dimension gain; it remains *larger* than the margin's curvature part.

B.3 The composite split-sample score

Write $\Psi(\gamma^A, \gamma^B) := 2\Theta(\frac{\gamma^A+\gamma^B}{2}) - \frac{1}{2}\Theta(\gamma^A) - \frac{1}{2}\Theta(\gamma^B)$. Differentiating in the first argument, $D_A\Psi = 2 \cdot \frac{1}{2}D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\gamma^A) = D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\gamma^A)$, which is (47). Evaluating along e_A and expanding both derivatives at γ_0,

$$\begin{aligned} D_A\Psi[e_A] &= D\Theta_0[e_A] + D^2\Theta[\bar{e}, e_A] - \tfrac{1}{2}D\Theta_0[e_A] - \tfrac{1}{2}D^2\Theta[e_A, e_A] + O(\text{3rd}) \\ &= \tfrac{1}{2}D\Theta_0[e_A] + \tfrac{1}{2}D^2\Theta[e_A, e_B] + O(\text{3rd}), \end{aligned}$$

using $D^2\Theta[\bar{e}, e_A] = \frac{1}{2}D^2\Theta[e_A, e_A] + \frac{1}{2}D^2\Theta[e_A, e_B]$. Adding the symmetric expression and subtracting from (46) gives (48). By contrast, the v1–v2 correction subtracts $\frac{1}{2}D\Theta(\gamma^A)[e_A] + \frac{1}{2}D\Theta(\gamma^B)[e_B] = \frac{1}{2}D\Theta_0[e_A + e_B] + \frac{1}{2}\{D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]\}$, whose second bracket has no counterpart in (46) and therefore survives with a minus sign. The difference between the two scores is $D\Theta(\bar{\gamma}) - D\Theta(\gamma^A) = -\frac{1}{2}D^2\Theta[\Delta, \cdot] + O(\|\Delta\|^2)$, i.e. exactly of the order the quadratic correction operates at — which is why the two look interchangeable at first order and are not.

B.4 What is not proved

The joint CLT for the two-band linear term follows from the Cramér–Wold device and the Lindeberg–Lyapunov condition of Assumption 5.1(g) applied to the summed per-unit influence $v_S^*(X_i, W_i)\epsilon_{S,i} + v_Y^*(X_i, W_i)\epsilon_{Y,i}$; the within-unit correlation ρ_{SY} is absorbed into σ_n^2 by construction (15). Beyond the order calculations of B.1–B.3, complete proofs of Theorems 4.6, 5.3 and 5.8 are not written. Four items are not merely mechanical, and each is stated as an assumption above rather than derived:

(1) **The second-derivative modulus (6).** Assumption 1.8(a)–(b), the third-order half of Assumption 5.1(f), and condition (iii) of Theorem 5.8 all rest on a Lipschitz-type bound for $\gamma \mapsto D^2\Theta(\gamma)$ that we have not established. Remark 4.8 converts each candidate modulus into a smoothness requirement: the one-gradient modulus gives $s > (3d-1)/4$, weaker than $(d+1)/2$ at $d = 2$; the two-gradient modulus gives $s > (3d+1)/4$; v3's crude $\|\bar{\Delta}\|_{C^1}^3$ gives $s > 3(d+1)/4$. This formulation is deliberately weaker than postulating $D^3\Theta$, which would require $\gamma_0 \in C^3$ and so contradict the parameter space for $s < 3$. The same bound would settle the jackknife gap of Remark 4.3. *This is the single most important open item in the paper.*

(2) **Uniform geometry along the path.** Assumption 4.1(c)–(d) — that the estimated level sets remain uniformly regular and transversal along $\bar{\gamma}+t\Delta$ — is assumed. For sieve first stages it should follow from $\|\hat{\gamma} - \gamma_0\|_{C^1} = o_p(1)$ and the implicit function theorem, uniformly; we have not written that argument, and it fails outright for non-smooth learners (Remark 4.4).

(3) **Band and Gram control.** Proposition 2.4(b)–(c) and Assumption 5.1(b),(d′) require an empirical-process argument for a K_n-dimensional band average and for $\hat{v}^*$ in the standard-deviation norm. This is the standard but nontrivial part of a sieve variance proof; we assume it.

(4) **Attainment of the margin bound.** Remark 3.4 exhibits mesh-resonant examples in which the margin diagonal bound $K_n^{1+1/d}/n$ is attained, and generic examples in which the integral is an order smaller. The exact order for curved margins, general d, and heteroskedastic errors is open; settling it would sharpen (or weaken) the $s \geq d$ of Theorem 2.2(i). Until then $K_n^{1+1/d}/n$ is used only as an upper bound.

(5) **Stochastic equicontinuity.** Assumption 1.8(c) is supplied by the global VC bound (8), which yields the window edge $a < d/(3d-2)$. A local-complexity argument — only the $O(\varkappa^{-(d-1)})$ basis functions meeting the margins are active — would plausibly improve it and relax that edge for $d \geq 3$. We have not attempted it.

(6) **Composite-score conditions for Theorem 5.8.** The representer conditions are stated for the differences $D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^j)$ rather than inherited from the ordinary case, and the ratio-equivalence of the composite-score variance estimator to (15) is asserted, not proved.

The corner parts of the argument, by contrast, require only the transversality bound, the dimension count $\dim \mathcal{C} = d - 2$, and Assumption 1.6(c) for the lower-bound direction.

C Numerical Verification of Theorem 3.1

Theorem 3.1 was checked against exact area computations in $d = 2$ on four designs, chosen so that between them every term of (18)–(19) is activated in isolation. In each case $\Theta(t) = \int \mathbb{1}\{g_0 + tu \geq 0\}\mathbb{1}\{h_0 + tv \geq 0\}w$ has a closed form; $\Theta''(0)$ is obtained by a Richardson-extrapolated central second difference and compared with the right-hand side of (18).

1. *Radially weighted disk.* $g_0 = 1 - \|x\|^2$, no second threshold, $w = \rho(\|x\|)$, $u \equiv a$. Only the transport and curvature terms of Q_g are active, and (19) reduces to $\pi a^2 \rho'(1)/2$. Relative error 6×10^{-10}.

2. *Disk with a non-constant direction.* $g_0 = 1 - \|x\|^2$, $w \equiv 1$, $u(x) = \|x\|^2$, so $\{g_t \geq 0\}$ is the disk of radius $(1-t)^{-1/2}$ and $\Theta(t) = \pi/(1-t)$. This activates the $u\,(n_g \cdot \nabla u)$ term. Formula and truth both give $D\Theta = \pi$, $D^2\Theta = 2\pi$ to ten digits.

3. *Disk cut by an oblique chord.* $g_0 = 1 - \|x\|^2$, $h_0 = x_1 - \kappa$, $w \equiv 1$, $u \equiv a$, $v \equiv c$. Both margin forms vanish, isolating the two corner diagonals and the corner cross term at $\cos\omega = -\kappa$. Checked at $\kappa \in \{0.6, 0.2, -0.5\}$ (both signs of $\cos\omega$) and $(a, c) \in \{(1, 1), (1.7, -0.9), (0.4, 2.1)\}$; maximum relative error 2×10^{-10}.

4. *Lens of two disks.* $g_0 = 1 - \|x - p\|^2$, $h_0 = 1 - \|x - q\|^2$ with $\|p - q\| = D$, $w \equiv 1$, $u \equiv a$, $v \equiv c$; both surfaces curved and obliquely intersecting, with $\cos\omega = 1 - D^2/2$. The prediction $\{ac - \frac{1}{2}(a^2 + c^2)\cos\omega\}/|\sin\omega|$ matches the exact lens area's second derivative at $D \in \{0.8, 1.2, 1.7\}$ and the same three (a, c) pairs; maximum relative error 2×10^{-9}.

Designs 3 and 4 are the substantive ones. Varying (a, c) over three non-proportional pairs identifies the quadratic form in (a, c) separately, so they confirm all three coefficients

$$\underbrace{\frac{2}{\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|}}_{\text{cross}}, \qquad \underbrace{\frac{-\cos\omega}{\|\nabla g_0\|^2\,|\sin\omega|}}_{Q_g \text{ diag.}}, \qquad \underbrace{\frac{-\cos\omega}{\|\nabla h_0\|^2\,|\sin\omega|}}_{Q_h \text{ diag.}},$$

and their relative signs, at both signs of $\cos\omega$.

The sharpest single check is design 4 with a *one-sided* perturbation, $v \equiv 0$: there the cross term is identically zero, so anything nonzero must come from the Q_g corner diagonal. At $D = 0.8, 1.2, 1.7$ ($\cos\omega = +0.680, +0.280, -0.445$) and $a = 1$ the exact lens curvatures are -0.463713017, -0.145833333, $+0.248456065$, against formula values -0.463713017, -0.145833333, $+0.248456064$ — including the sign reversal as $\cos\omega$ crosses zero. This is the term that v1's order discussion omitted and that (27) restores: the corner contributes to the plug-in bias even when the two contrast errors are uncorrelated, provided the margins do not meet orthogonally.

5. Polarization and mesh resonance. Two further checks support statements made in the body.

(a) The polarization identity (37). On all thirteen (D or κ, a, c) configurations of designs 3 and 4, the numerically differenced $q_{\text{full}} - q_g - q_h$ agrees with $2C[a, c]$ to full double precision, while the sum of the two $\cos\omega$ diagonals — which is nonzero and often larger in magnitude — is absorbed entirely into q_g and q_h. For instance at $D = 1.2$, $a = c = 1$: $q_{\text{full}} = 0.750000$, $q_g = q_h = -0.145833$, so $q_{\text{full}} - q_g - q_h = 1.041667 = ac/|\sin\omega| = 2C$, against a corner-diagonal sum of -0.291667. This is what licenses Proposition 4.10: margins-only debiasing removes the $\cos\omega$ diagonals and leaves the cross term.

(b) Mesh resonance in (29). With periodic uniform cubic tensor B-splines on the torus in $d = 2$ ($K_n = m^2$, mesh $\varkappa = 1/m$), we computed $\mathcal{V} = \mathcal{V}_1 \otimes \mathcal{V}_1$ in closed form — the diagnostic $\int_0^1 \mathcal{V}_1 = m$ holds exactly and $\|\mathcal{V}_1'\|_\infty = 1.226\, m^2$ to four digits for $m = 16, 32, 64$, confirming the pointwise scalings (22) — and evaluated $\int_{\mathcal{M}_g} \psi\, \partial_n \mathcal{V}$ along straight margins of slope α for $m = 16, 32, 64, 128$. Reporting $|I|/K_n$ (flat $\Rightarrow I \asymp K_n$; growing linearly in $m \Rightarrow I \asymp K_n^{1+1/d}$):

α	$m = 16$	32	64	128
0 (mesh-aligned)	16.1	20.7	66.6	27.4
1 (resonant)	3.6	5.2	15.4	6.9
$1/2$, $8/5$ (rational, $q \geq 2$)	≤ 0.12	≤ 0.20	≤ 0.09	≤ 0.59
$1/\pi$, $\sqrt{2}$, $\sqrt{5} - 1$ (irrational)	≤ 0.43	≤ 0.65	≤ 0.40	≤ 0.48

The first two rows fluctuate in phase but grow with m (so $I \asymp K_n^{1+1/d}$); the last two are bounded (so $I \asymp K_n$). The resulting hundredfold spread at $m = 128$ is the evidence for Remark 3.4.

6. The support-edge term of Remark 1.10. Take the unit disk $g_0 = 1 - \|x\|^2$ with the *fixed* support $\mathcal{X} = \{x_1 \geq \kappa\}$, $w \equiv 1$, $u \equiv a$, and no second threshold. The exact curvature of the circular-segment area matches $-2a^2 \cos\omega_\partial/(4\,|\sin\omega_\partial|)$ with $\cos\omega_\partial = -\kappa$ at $\kappa \in \{0.6, 0.2, 0, -0.5\}$ and $a \in \{1, 1.7\}$ to ten digits, including the exact vanishing at $\kappa = 0$ (orthogonal meeting). This confirms both the form of the edge term and that it carries no cross component.

7. The doubly debiased score (Remark 5.6). On the two-disk lens with $D = 1.2$ and direction $\eta = (a, c) = (1, 0.8)$, write $\phi(t) := \Theta(\gamma_0 + t\eta)$, so $\phi''(0) = D^2\Theta[\eta, \eta] = +0.5942$, and set $e_j = t_j\eta$. Three estimators are compared: Ψ (pure SS), the v1–v2 estimator (45), and the composite-score estimator (49), all evaluated exactly.

(a) The counterexample $e_A = e_B = b$. Ψ reduces to $\Theta(\gamma_0 + b)$ and (45) has error -2.679×10^{-3}, -7.044×10^{-4}, -1.808×10^{-4}, -4.580×10^{-5} at $t = 0.1, 0.05, 0.025, 0.0125$ — quartering as t halves, i.e. $O(t^2)$, and matching $-t^2\phi''(0)/2$ $(-2.971\times10^{-3}, \dots, -4.642\times10^{-5})$ to the accuracy of the third-order term. The error is quadratic, not cubic.

(b) Independent mean-zero errors. With t_A, t_B i.i.d. $\mathcal{N}(0, \varsigma^2)$ and 4×10^5 draws:

ς	$\mathbb{E}[\Psi - \Theta_0]$	$\mathbb{E}[(45) - \Theta_0]$	$\mathbb{E}[(49) - \Theta_0]$	$-\varsigma^2\phi''(0)$
0.08	3.4×10^{-5}	-3.86×10^{-3}	1.7×10^{-5}	-3.80×10^{-3}
0.04	-1.0×10^{-4}	-9.56×10^{-4}	2.1×10^{-6}	-9.51×10^{-4}
0.02	3.0×10^{-5}	-2.38×10^{-4}	1.7×10^{-7}	-2.38×10^{-4}
0.01	-3.9×10^{-6}	-5.96×10^{-5}	1.2×10^{-9}	-5.94×10^{-5}

Ψ is unbiased at this order (its leftover $\frac{1}{2}D^2\Theta[e_A, e_B]$ is mean zero); (45) has bias $-\varsigma^2\phi''(0)$ to three significant digits, confirming Remark 5.6; and (49) is unbiased with a residual decaying like ς^4.

8. The two polarization families (Remark 4.5). On the lens design with $D = 1.2$, $\eta = (a, c) = (1, 0.8)$ and half-sample errors $e_j = t_j\eta$, all four objects of (34) and (35) are computable in closed form. The two *within*-family identities $\hat{\Theta}_{SS}^{\text{mar}} - \text{corner} - \hat{\Theta}_{SS}$ and $\hat{\Theta}_{SS}^{\text{mar,jack}} - \widehat{\text{corner}} - \hat{\Theta}_{SS}^{\text{jack}}$ evaluate to *exactly* 0 (machine zero) at every (t_A, t_B) tried. The *cross*-family gap $\hat{\Theta}_{SS}^{\text{jack}} - \hat{\Theta}_{SS}$, which v3 asserted was zero, is not:

t_A	t_B	$\hat{\Theta}_{SS}^{\text{jack}} - \hat{\Theta}_{SS}$	$-\phi^{(4)}/384$	ratio
0.40	0.16	-8.943×10^{-6}	-8.896×10^{-6}	1.01
0.20	0.08	-8.674×10^{-7}	-8.646×10^{-7}	1.00

confirming Lemma 4.2's constant $1/384$ to one percent. (Smaller t_j are not reported: the five-point stencil for $\phi^{(4)}$ amplifies rounding by h^{-4} and the comparison becomes noise-limited below $\sim 10^{-7}$.) This is the numerical content of Remark 4.5: each family is internally exact, and only the cross-family statement carries the fourth-order error.

The scripts for items 1–8 sit alongside this document:

`verify_D2_theorem.py`, `verify_polarization.py`, `verify_mesh_resonance.py`, `verify_dd_score.py`, `verify_offdiag_order.py`, `verify_two_families.py`.

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